Statistical Analysis of Stop and Search Practices on Black Communities in the UK

Michel Mesquita, Ph.D., CStat

October 13, 2026

Table of contents

1 Statistical Analysis of UK Stop and Search **5**
- 1.1 Research Questions . . . 5
- 1.2 Scope and Data . . . 5
- 1.3 Key Findings . . . 6

2 Introduction **7**

3 Literature Review **9**

4 Methodology **13**
- 4.1 Data Sources . . . 13
- 4.2 Data Cleaning Steps . . . 14
- 4.3 Data Processing and Filtering . . . 14
  - 4.3.1 Stop and Search Data . . . 14
  - 4.3.2 Census Data . . . 15
  - 4.3.3 Merging and Aggregation . . . 15
  - 4.3.4 Additional Filtering in Derived Analyses . . . 15
  - 4.3.5 Reproducibility . . . 16
- 4.4 The Pooled Model . . . 16
- 4.5 Quasi-Poisson: Justification . . . 17
- 4.6 Per-Year IRR Models . . . 17
- 4.7 Random Forest Model as a Complementary Analysis . . . 17
  - 4.7.1 Rationale for Using Random Forest in Addition to the Main Model . . . 17
  - 4.7.2 Overview of the Random Forest Model . . . 18
  - 4.7.3 Interpretation of Random Forest Output . . . 19

5 Results **20**
- 5.1 Census data . . . 20
  - 5.1.1 Ethnic Composition Across English Regions . . . 20
  - 5.1.2 Population Distribution by Ethnic Group and Region . . . 21
  - 5.1.3 Geographic Distribution of the Black or Black British Population . . . 21
- 5.2 Stop and Search data . . . 22
  - 5.2.1 Total Stop and Searches by Year . . . 22
  - 5.2.2 Disparity Rates Relative to White Individuals . . . 24
  - 5.2.3 Outcome and Link . . . 24
- 5.3 Modelling . . . 25
  - 5.3.1 **Overview of Models Tested** . . . 25
  - 5.3.2 **Key Considerations** . . . 26

5.4 Chosen Model . . . 26
5.4.1 Overview of the Poisson Regression . . . 26
5.4.2 Model results . . . 27
5.4.3 Model Coefficients: Interpretation . . . 28
5.4.4 Model Fit and Diagnostics . . . 29
5.4.5 Addressing the Project Questions . . . 30
5.4.6 Implications of Findings . . . 31
5.4.7 Policy Implications . . . 32
5.5 Random Forest Model: Machine Learning Approach . . . 33
5.5.1 Variable Importance and Interpretation . . . 33
5.5.2 Implications for Project Objectives . . . 34
5.6 Five-Year Disparity Trend . . . 35
5.7 Regional Force Rankings . . . 36
5.8 Outcome Disparity by Ethnicity . . . 37
5.9 Age by Ethnicity Interaction . . . 38
5.10 Racial Skew by Search Reason . . . 39
5.11 Missing Ethnicity Data: Systematic Patterns . . . 40

6 Discussion Section **43**
6.1 Interpretation of Results and Comparison with Government Findings . . . 43
6.2 Contrasting with Vomfell and Stewart's Findings on Officer Bias and Over-Patrolling . . . 44
6.3 Implications for Policy and Policing . . . 44
6.4 Is the Disparity Getting Worse? . . . 45
6.5 The South West Anomaly . . . 45
6.6 Outcome Disparity and Differential Suspicion Thresholds . . . 46
6.7 Disparity Is Not Explained by Search Reason . . . 46
6.8 Missing Ethnicity Data: Analytical Implications . . . 47
6.9 Limitations: Missing Data and Census Denominator Lag . . . 47

7 Conclusion **49**
7.1 Key Findings . . . 49
7.2 **Evidence-Based Implications** . . . 50
7.3 **Closing Statement** . . . 51

8 References **52**

Appendices **54**

A Model Diagnostics **54**
A.1 Pearson Residuals vs Fitted Values . . . 54
A.2 Normal Q-Q Plot of Pearson Residuals . . . 55
A.3 Cook's Distance . . . 56
A.4 Observed vs Expected Counts . . . 57
A.5 Summary of Diagnostic Statistics . . . 58

B Sensitivity Analysis **59**
B.1 Leave-One-Covariate-Out 59
B.2 Bootstrap Confidence Intervals 60
B.3 Poisson vs Quasi-Poisson Standard Errors 60
B.4 Finer Population Denominators 63

C Inverse Propensity Weighted Estimates **65**
C.1 Method 65
C.2 Weight Distributions 66
C.3 GLM vs IPW Comparison 67
C.4 Interpretation and Limitations 69

1 Statistical Analysis of UK Stop and Search

> **Note**
>
> **Revision, October 2026.** This version adds a sensitivity analysis using Census population by sex and age band (Section B.4), and clarifies the description of the population offset and of the change in the adjusted disparity between 2020/21 and 2024/25. The main estimates are unchanged.

This report presents the final findings of the **UNJUST** project, which examines the impact of police stop and search powers on Black communities in England and Wales. Conducted as part of the **Statisticians for Society** programme, the project delivers a comprehensive analysis supported by literature reviews, statistical modelling, and data evaluation. All methodologies and findings are documented and accessible through the GitHub repository accompanying this report (Mesquita, 2023).

1.1 Research Questions

The analysis addresses four research questions identified by **UNJUST**:

1. **Disproportionate Targeting**: Do Black individuals in metropolitan areas experience higher rates of drug-related stop and searches compared to other ethnic groups?
2. **Youth Impact**: How do stop and search rates vary for young individuals (under 25 years old) in drug-related cases?
3. **Trend over Time**: Has the adjusted disparity in stop and search rates changed over the five-year period 2020/21–2024/25?
4. **Outcome Disparity**: Do post-search outcomes differ by ethnicity, suggesting differential thresholds for suspicion?

1.2 Scope and Data

This report covers five complete financial years, 2020/21 through 2024/25, using the GOV.UK Stop and Search Open Data (556,000+ aggregated records representing approximately 2.9 million individual stop-and-search incidents) merged with ONS Census 2021 population denominators.

1.3 Key Findings

- **Raw national disparity**: In 2024/25, Black individuals were stopped at **3.6 times** the rate of White individuals, the headline figure shown on the accompanying dashboard. This is the unadjusted rate ratio (rounded to 1 d.p.), calculated directly from search counts divided by 2021 Census population denominators, with no covariates controlled.
- **Persistent adjusted disparity**: From a model that includes region, sex, age band and reason for search, with population by region and ethnic group as the offset, the adjusted incidence rate ratio (IRR) is **3× in 2020/21 and 3.3× in 2024/25** (see Section 5.6). The adjusted figure is lower than the raw 3.6× because the model compares ethnic groups within the same region; the gap between the two measures reflects regional composition, not measurement error.
- **Interaction effects**: The compounding of ethnicity, search reason, and geography means that a Black individual stopped for drug-related reasons in London is searched at approximately **21.9 times** the rate of a White individual stopped for non-drug reasons in the East Midlands, holding sex and age constant.
- **Regional anomaly**: The South West has the highest Black/White disparity ratio (5.1×), exceeding London (3.6× pooled across all five years), suggesting nationally uneven patterns of disproportionality.
- **Outcome disparity**: Black individuals have one of the highest arrest rates across all ethnic groups (16%, tied with Mixed), 2.7 percentage points above White individuals (13.3%), yet similar No Further Action rates (~69%), raising questions about differential suspicion thresholds.
- **Disparity cuts across all search reasons**: The racial gap is not confined to drug stops. Every major reason for search, from weapons to articles for use in crime, shows a Black/White rate ratio substantially above parity, indicating a systemic rather than enforcement-specific pattern (see Section 5.10).
- **Missing ethnicity data is concentrated in a subset of forces**: Self-defined ethnicity is voluntary: individuals are not required to state it during a stop, and around 1 in 10 records nationally has no ethnicity recorded. The rate varies substantially across forces, which limits the comparability of force-level disparity estimates (see Section 5.11).

These findings are intended to inform policy reform discussions and contribute to broader conversations on fairness and equity in law enforcement. An interactive dashboard accompanying this report is available at unjust.netlify.app.

2 Introduction

Stop and search is a key aspect of policing in the United Kingdom, involving efforts to identify and prevent criminal activity. Previous studies indicate that ethnic minorities, particularly Black individuals, are subject to stop and search procedures at higher rates compared to other groups (Home Office (GOV.UK), 2023; IER UK, 2023; Commission on Race and Ethnic Disparities, 2021; Vomfell and Stewart, 2021; Social Justice, 2020; Equality and Commission, 2010). The UNJUST project, initiated under the Statisticians for Society program (Royal Statistical Society (RSS), 2023), aims to provide a detailed, data-driven examination of these patterns to enhance understanding and inform policy discussions.

The project addresses four research questions identified by UNJUST:

1. **Disproportionate Targeting**: Do Black individuals in metropolitan areas experience higher rates of drug-related stop and searches compared to other ethnic groups?
2. **Youth Impact**: How do stop and search rates vary for young individuals (under 25 years old) in drug-related cases?
3. **Trend over Time**: Has the adjusted disparity in stop and search rates changed over the five-year period 2020/21–2024/25?
4. **Outcome Disparity**: Do post-search outcomes differ by ethnicity, suggesting differential thresholds for suspicion?

These questions align with broader concerns about systemic inequalities within policing practices and their potential impact on community trust and social justice.

To address these questions, the UNJUST project employs comprehensive data analysis and statistical modelling, utilising techniques that offer insights into the distribution of stop and search incidents across different ethnic groups. This report builds on the foundational work of researchers such as Vomfell and Stewart (2021), whose studies highlight the combined impact of officer discretion and the increased patrolling of ethnic minority areas. By analysing stop and search data from five complete financial years, 2020/21 to 2024/25, covering 556,000+ aggregated records representing approximately 2.9 million individual stop-and-search incidents, the project provides an updated perspective on the prevalence of these disparities and, crucially, whether they are improving or worsening over time.

One notable development in this area is the increasing use of artificial intelligence (AI) to support police operations. AI-driven tools, such as Live Facial Recognition (LFR), have been trialled by the Metropolitan Police in London to enhance stop and search methods. This technology enables real-time identification of individuals on police watchlists by scanning and recognising faces from distances of 10 to 20 metres. The system compares faces against databases containing thousands of known individuals, and when a match is detected, officers are alerted to potentially stop the individual for further questioning. According to recent

reports, AI has the potential to improve the efficiency and objectivity of stop and search practices, contributing to more data-driven policing methods (Sylvester, 2024). While the UNJUST project does not incorporate AI-based stop and search data in its current analysis, understanding the role of emerging technologies is crucial for future efforts aimed at reducing disparities and enhancing the fairness of policing practices.

The findings presented in this report offer a rigorous, evidence-based analysis of the current state of stop and search practices, with a particular focus on racial disparities. By examining these issues through the lens of statistical rigour, the UNJUST project aims to contribute valuable insights to ongoing discussions about justice, equality, and effective law enforcement.

3 Literature Review

Stop and search practices by the police in the United Kingdom are intended to play a significant role in maintaining public order and safety. These powers, primarily governed by the Police and Criminal Evidence Act 1984 (PACE) (legislation.gov.uk, 1984) and the Criminal Justice and Public Order Act 1994 (CJPOA) (legislation.gov.uk, 1994), enable the police to intervene and prevent potential crimes, particularly in areas with higher levels of criminal activity. However, concerns have been raised regarding the disproportionate use of stop and search on Black and ethnic minority communities. This review synthesizes key findings from recent literature, exploring both the effectiveness of stop and search as a policing tool and the concerns over racial disparities.

Stop and search powers have evolved over time and have been utilized in crime prevention efforts. For instance, data from the Home Office for the year ending March 2023 indicates that 547,002 stop and searches were conducted in England and Wales, marking a 3% increase from the previous year. Of these searches, 14% resulted in arrests, highlighting their role in identifying potential offenders and intervening in criminal behaviour (Home Office (GOV.UK), 2023). However, disparities persist, as Black individuals were 5.5 times more likely to be stopped than White individuals in 2020/21, with this ratio decreasing to 4.8 in 2021/22 and 4.1 in 2022/23. These figures prompt further examination of the factors underlying such disparities (Home Office (GOV.UK), 2023).

Broader context on ethnic disparities across the criminal justice system is provided by the Ministry of Justice's annual *Statistics on Race and the Criminal Justice System* (Ministry of Justice, 2023). The most recent edition documents persistent over-representation of Black individuals at every stage of the criminal justice pipeline: from stop and search and arrests through to prosecution, conviction, and custodial sentencing. Notably, Black individuals accounted for approximately 9% of arrests in England and Wales despite comprising around 3.5% of the population, a pattern of disproportionality that closely mirrors the disparities observed in stop and search data. This consistency across multiple points of contact with the criminal justice system reinforces the argument that the disparities measured in stop and search are not statistical artefacts of enforcement discretion alone, but form part of a broader, systemic pattern of unequal treatment (Ministry of Justice, 2023).

The causes of these disparities are multifaceted. Racial disparities in stop and search rates have been consistently observed, with several underlying factors potentially contributing to this trend. These include the demographic concentration of certain ethnic groups in areas with higher crime rates and operational strategies focused on specific high-risk areas. Localized crime trends, patterns of criminal activity occurring within specific geographic areas, often guide such strategies, with increased law enforcement activity targeting regions with higher reported rates of violent crime or drug-related offenses. Additionally, the

subjective nature of suspicion during searches adds another layer of complexity. For example, the IER UK report on York highlights that Black individuals in North Yorkshire are 18 times more likely to be stopped than White individuals, partly attributed to over-policing in neighborhoods with higher crime rates (IER UK, 2023).

These observed disparities highlight not only operational challenges in stop and search but also raise critical questions about broader systemic inequalities that influence its application. Addressing these inequalities requires a deeper understanding of their socio-economic and historical roots, as explored in reports like the Runnymede Trust's The Colour of Money (Runnymede, 2020) and the Equality and Human Rights Commission's (EHRC) Stop and Think (Equality and Commission, 2010).

Stop and search is one of many tools used by the police to tackle crime, but its effectiveness and broader implications remain contested. The Runnymede Trust's The Colour of Money report does not directly address policing tactics like stop and search but highlights the systemic racial inequalities that influence socio-economic outcomes and the lived experiences of Black and ethnic minority communities. These systemic factors, including discrimination and socio-economic disparities, may indirectly shape the context in which stop and search is applied and perceived. The report underscores that addressing such inequalities requires systemic changes, suggesting that targeted interventions in high-crime areas may not fully resolve underlying racial disparities without broader structural reforms to tackle economic and social inequities (Runnymede, 2020).

Similarly, the Equality and Human Rights Commission's (EHRC) report, Stop and Think, critically examines the disproportionate use of stop and search powers (Equality and Commission, 2010). It highlights that while stop and search can provide a means to confirm or allay suspicion without arrest, the evidence suggests it has limited effectiveness in reducing crime rates, contributing only to a 0.2% reduction in disruptable crimes. The report also underscores that disproportionate targeting of Black and Asian individuals undermines trust in the police, negatively affecting community relations. The EHRC emphasizes that any effectiveness of stop and search should be balanced against these societal costs, suggesting that policing strategies must be both fair and proportionate to foster public confidence.

Concerns have been raised about the broader societal impacts of stop and search, particularly regarding its effect on community trust. Repeated law enforcement encounters can erode trust, fostering feelings of alienation and mistrust among minority communities. The use of facial recognition technology adds another layer of complexity (Sylvester, 2024), as advanced surveillance tools may reinforce perceptions of unfair targeting. While intended to enhance policing efficiency, facial recognition technology raises concerns about algorithmic biases that disproportionately misidentify individuals from minority groups, further compounding these perceptions. Such tools, when integrated into stop and search operations, may exacerbate perceptions of bias, particularly if algorithmic inaccuracies disproportionately affect minority groups. Nevertheless, the use of such technologies is arguably framed as a means to enhance data-driven decision-making and refine localized risk assessments to address specific crime trends.

Conversely, police forces emphasize that stop and search is used proportionately and in response to specific intelligence or local crime trends. According to the Commission on Race

and Ethnic Disparities, operational decisions related to stop and search are often guided by crime data and localized risk assessments, which aim to address high-risk areas while maintaining fairness and transparency (Commission on Race and Ethnic Disparities, 2021).

The Centre for Social Justice's Facing the Facts report (2020) highlights the importance of understanding socio-economic disparities both within and across ethnic groups. It challenges the broad use of the "BAME" (Black and Minority Ethnic) category, arguing that granular data, detailed and disaggregated information on individual ethnic subgroups, provides a more nuanced understanding of disparities in areas such as education, employment, and family stability (Social Justice, 2020). These disparities often overlap with patterns of localized crime, influencing both the prevalence of police interventions and public perceptions of fairness in policing practices. By addressing these socio-economic disparities and fostering inclusivity across public services, these strategies aim to reduce systemic drivers of racial disparities in policing and enhance community trust. Such interventions not only address root causes of crime but may also reduce the need for targeted policing strategies that disproportionately affect minority communities. In tandem with policing reforms, addressing broader socio-economic inequalities through targeted interventions in education and employment could reduce the systemic drivers of these disparities.

The Home Office's report for 2023 indicates that 71% of stop and searches resulted in no further action, raising questions about the criteria for suspicion and the balance between proactive policing and the protection of civil liberties (Home Office (GOV.UK), 2023). The Home Office data underscores the scale of stop and search operations, yet the high proportion of no-action outcomes highlights the challenges of balancing proactive policing with civil liberties. While some may interpret this as a sign of inefficiency, it can also be seen as indicative of the difficulty in balancing public safety with the protection of civil liberties.

In response to these concerns, several recommendations have been proposed. The Commission on Race and Ethnic Disparities advocates for greater transparency and improved community relations, suggesting that community-led policing models and more rigorous oversight could help restore trust between police and minority communities (Commission on Race and Ethnic Disparities, 2021). Additionally, the IER UK's anti-racism strategy for York emphasizes the importance of anti-racism training for public service professionals, including cultural awareness, unconscious bias, and inclusivity practices. The strategy focuses on fostering improved relations with York's diverse communities by embedding anti-racism principles across all public services. It advocates for greater community engagement, encouraging active dialogue between public institutions and underrepresented groups to shape policies collaboratively. Furthermore, the strategy highlights the need for accountability through regular monitoring and transparent reporting on racial disparities, ensuring that systemic reforms lead to measurable progress (IER UK, 2023).

In conclusion, while stop and search remains a vital tool for maintaining public safety, evidence highlights its disproportionate impact on Black and ethnic minority communities, warranting careful consideration. These disparities stem from both operational and systemic challenges, including socio-economic inequalities, subjective enforcement criteria, shaped by an officer's interpretation of 'reasonable suspicion', and broader societal impacts on community trust. Subjective enforcement, influenced by implicit biases and contextual judgments, exacerbates these disparities, perpetuating systemic inequalities. Achieving

equitable and effective policing requires addressing not only operational practices but also the socio-economic and systemic factors that underpin racial disparities, ensuring a holistic approach to reform. Reform efforts must address these intersecting concerns to strike a balance between effective crime prevention and the protection of civil liberties. Future initiatives, such as the UNJUST project, will be instrumental in providing evidence-based insights to guide policy reforms, ensuring that stop and search practices are equitable, transparent, and conducive to public trust.

4 Methodology

4.1 Data Sources

The analysis utilizes two primary data sources:

1. **GOV.UK Stop and Search Open Data (2020/21–2024/25)**: This dataset provides comprehensive records of stop and search incidents across England and Wales, including variables such as ethnicity, age, sex, region, and reason for search. The five-year extract covers the financial years ending March 2021 through March 2025, comprising 556,000+ aggregated records representing approximately 2.9 million individual stop-and-search incidents after cleaning. The data are released by the Home Office (Home Office (GOV.UK), 2024) and include all searches conducted under Section 1 of the Police and Criminal Evidence Act (PACE) and Section 60 of the Criminal Justice and Public Order Act (CJPOA).
2. **ONS Census 2021**: Used to obtain population denominators by ethnicity and region, serving as the offset in the rate models (National Statistics (ONS), 2022). We assume that the 2021 population counts remain a reasonable approximation across all five financial years, a limitation discussed in the Discussion chapter.

The analysis is inspired by Vomfell and Stewart (2021), who identified ethnic disparities in stop and search in the United Kingdom.

The selected variables, their code names, and data types are: number of stop and search events (`number_of_searches`, continuous), year (`financial_year`, categorical with five levels), region name (`region`, categorical with ten levels), ethnicity group defined by the person stopped (`self_defined_ethnicity_group`, categorical), sex (`sex`, categorical), age group (`age_group`, categorical), reason for search (`reason_for_search`, categorical), and population from census data (`population`, continuous).

Post-processing of the stop and search data includes: (1) verifying totals match published government figures ((GOV.UK), 2024); (2) filtering to Section 1 (PACE) and Section 60 (CJPOA) only; (3) converting categorical variables to factors; (4) removing vehicle searches (`N/A - vehicle search`), "Unknown" age group, "Not Stated" ethnicity, "Unknown" sex, and region `0` (British Transport Police, excluded because the 2021 Census provides no BTP-specific denominator); (5) converting `age_group` to a binary variable where `1` = under 25 and `0` = 25 or above; and (6) converting `reason_for_search` to a binary variable where `1` = Drugs and `0` = otherwise.

4.2 Data Cleaning Steps

- **Removal of Incomplete Cases**: Excluded entries with missing or irrelevant data (e.g., vehicle searches).
- **Categorization**: Transformed continuous variables into categorical factors where appropriate.
- **Legislative Filtering**: Ensured that all stop and search incidents analyzed were conducted under Section 1 (PACE) or Section 60 (CJPOA).

4.3 Data Processing and Filtering

The following steps describe how the raw data were processed and filtered prior to analysis. All operations are implemented in `01_load_and_clean.R` and are fully reproducible from the raw files.

4.3.1 Stop and Search Data

The raw dataset is the Home Office stop and search open data tables (March 2021 to March 2025 release), distributed as a single ODS file covering five financial years (2020/21–2024/25). Processing proceeded as follows:

1. **Vehicle searches removed.** Rows where `self_defined_ethnicity_group` equals "N/A - vehicle search" were excluded because these records lack person-level demographics.
2. **British Transport Police removed.** Records coded as region `0` (British Transport Police) were excluded because the 2021 Census provides no BTP-specific population denominator, making rate calculations impossible.
3. **Ethnicity harmonisation.** The five Home Office ethnicity groups (Asian or Asian British, Black or Black British, Mixed, Other Ethnic Group, White) were retained and mapped to short labels (Asian, Black, Mixed, Other, White). Records where `self_defined_ethnicity_group` is "Not Stated" were set to `NA` and excluded from disparity calculations but retained for total-count comparisons and missing-data analyses.
4. **Age binarisation.** The original age groups (Under 10, 10–17, 18–24, 25–29, 30 or over) were collapsed into a binary variable: `age_under25 = TRUE` for Under 10, 10–17, and 18–24; `FALSE` for 25–29 and 30 or over. Records with "Unknown" age were set to `NA`.
5. **Reason binarisation.** `reason_for_search` was converted to a binary indicator: `reason_drugs = TRUE` if the reason is "Drugs", `FALSE` otherwise. The original reason variable was retained for the reason-by-reason analyses in Section 5.10.

6. **Region name standardisation.** Two region names were recoded to match the ONS Census geography: "Eastern" → "East of England"; "Yorkshire and the Humber" → "Yorkshire and The Humber".
7. **Sex filtering.** Records with sex recorded as "Unknown" were excluded. Only "Male" and "Female" were retained.
8. **Legislative scope.** The raw data include searches under Section 1 (PACE) and Section 60 (CJPOA) only, as released by the Home Office. No additional legislative filtering was required.

4.3.2 Census Data

Population denominators were drawn from the ONS 2021 Census table on population by ethnicity and region (National Statistics (ONS), 2022). Processing steps:

1. **Ethnicity type filtered.** Only rows with `ethnicity_type = "ONS 2021 5+1"` (the five main groups plus Other) were retained.
2. **National aggregate excluded.** The "All - England And Wales" geography row was removed; only regional totals were kept.
3. **Five ethnic groups retained.** Asian, Black, White, Mixed, and Other, matching the stop and search categories.

4.3.3 Merging and Aggregation

Stop and search records were aggregated by financial year, region, and ethnicity (after excluding rows with missing ethnicity), then joined to the census population by region and ethnicity. This produced a merged dataset of stratified counts with population denominators, from which per-1,000 rates and model offsets were calculated.

For the Poisson regression models, the data were further stratified by sex, age group (binary), and reason for search (binary), producing the full factorial modelling dataset.

4.3.4 Additional Filtering in Derived Analyses

- **Reason-by-reason disparity** (Section 5.10): Black/White rate ratios were computed per search reason. Reasons with fewer than 1,000 total searches across all years and ethnicities were excluded as unreliable.
- **Missing ethnicity analysis** (Section 5.11): Force-level missingness was computed from the full dataset (including `NA` ethnicity records). Only forces with data in at least three of the five years were included in the persistent-missingness ranking.
- **NFA and outcome analysis**: Computed from all records with non-missing ethnicity, without further exclusions.

4.3.5 Reproducibility

All scripts are available in the project repository (`analysis/01_load_and_clean.R` through `analysis/06_export_for_dashboard.R`). Raw data can be downloaded via `data/download_data.sh`, which retrieves the Home Office ODS file and ONS census CSV from their published URLs. Running the scripts in numerical order from the repository root will reproduce all processed datasets and analysis outputs.

4.4 The Pooled Model

We model total searches as the dependent variable (Y) with region, ethnic group, sex, age group, and reason for search as independent variables. Because the raw count data exhibited severe overdispersion (estimated dispersion parameter $\hat{\phi} = 181.8$; see Section 4.5), we use a **Quasi-Poisson generalised linear model** with a log link and the census population by region and ethnic group as an offset. Every modelling cell within a region and ethnic group shares that population; a sensitivity analysis with population also split by sex and age band is reported in Section B.4.

Let Y be the number of searches observed for population stratum N. The mean is $\mu = N\exp(\mathbf{X}^T\beta)$, with linear predictor $\eta = \mathbf{X}^T\beta + \log N$. For the i-th stratum:

$$\mu_i = n_i \exp(x_i^T\beta) = \exp(x_i^T\beta + \log n_i) \tag{1}$$

using the notation in Davison (2009). Expanding the linear predictor over the five-year period:

$$\begin{aligned}\log(\mu_i) = \log(pop_i) + \beta_0 + \sum_{h=2}^{5}\beta_{1h}I(\text{year}_{ih}) + \sum_{l=1}^{L}\beta_{3l}I(\text{region}_{il}) + \sum_{j=1}^{J}\beta_{4j}I(\text{ethnic}_{ij}) \\ + \beta_5 I(\text{sex}_{i,\text{Male}}) + \beta_7 I(\text{age}_{i,<25}) + \beta_8 I(\text{drugs}_{i1})\end{aligned} \tag{2}$$

where $I(\cdot)$ is an indicator variable. Reference levels are: financial year **2020/21**, region **East Midlands**, ethnicity **White**, sex **Female**, age group **25 and above**, and reason **non-drug-related**.

Variable descriptions:

- pop_i: census population count for stratum i (National Statistics (ONS), 2022)
- $I(\text{year}_{ih})$ (`financial_year`): five levels: 2020/21, 2021/22, 2022/23, 2023/24, 2024/25
- $I(\text{region}_{il})$ (`region`): ten regions in England and Wales (East Midlands as reference)
- $I(\text{ethnic}_{ij})$ (`self_defined_ethnicity_group`): five groups: Asian or Asian British, Black or Black British, Mixed, Other Ethnic Group, White (reference)
- $I(\text{sex}_{i,\text{Male}})$ (`sex`): Male vs Female (reference)
- $I(\text{age}_{i,<25})$ (`age_group`): binary: 1 if under 25, 0 otherwise
- $I(\text{drugs}_{i1})$ (`reason_for_search`): binary: 1 if drug-related, 0 otherwise

4.5 Quasi-Poisson: Justification

A standard Poisson model assumes that the mean equals the variance ($\text{Var}(Y) = \mu$). In practice, count data on policing activity routinely exhibit **overdispersion**, variance substantially exceeding the mean, due to unmodelled heterogeneity across police forces, local crime environments, and seasonal patterns.

Fitting the pooled model with a standard Poisson family yields a residual deviance of 342,749 on 1,950 degrees of freedom, producing an estimated dispersion parameter of:

$$\hat{\phi} = \frac{\text{Residual deviance}}{\text{df}} = \frac{342{,}749}{1{,}950} \approx 181.8$$

This is far greater than 1, confirming severe overdispersion. A Quasi-Poisson model retains the same coefficient estimates as Poisson but scales all standard errors by $\sqrt{\hat{\phi}}$, producing correct (wider) confidence intervals and conservative inference. Using a Poisson family without this correction would dramatically understate uncertainty and inflate the apparent precision of the IRR estimates.

Note that Quasi-Poisson models do not yield a finite AIC; model adequacy is instead assessed via the pseudo-R^2 (Cohen, 1988) and residual deviance diagnostics. Full regression diagnostics, including Pearson residual plots, Cook's distance analysis, and an observed-vs-expected comparison, are presented in Appendix A. Sensitivity of the ethnicity IRRs to alternative model specifications is assessed in Appendix B, and a comparison with inverse propensity weighted estimates is provided in Appendix C.

4.6 Per-Year IRR Models

To examine whether the adjusted disparity in stop and search rates has changed over time (Research Question 3), we fit five separate Quasi-Poisson models, one per financial year, each using the same specification as the pooled model but without the `financial_year` term. This yields year-specific incidence rate ratios (IRRs) and 95% confidence intervals for each ethnic group relative to White individuals. The per-year Black IRRs are presented in the Results chapter (Section 5.6).

4.7 Random Forest Model as a Complementary Analysis

4.7.1 Rationale for Using Random Forest in Addition to the Main Model

The main model for this analysis employs a regression approach to examine the impact of various demographic and contextual factors on stop-and-search practices. While regression models are powerful for estimating specific associations and providing clear interpretations through coefficients, they have some limitations. Specifically, regression models assume

linearity and often require predefined interactions between predictors to fully capture complex relationships within the data. In contrast, random forests are a non-parametric machine learning approach that can automatically capture both non-linear relationships and interactions among variables without needing explicit specification. This capability makes random forests an ideal complementary tool for exploring complex, multi-dimensional patterns within stop-and-search data.

Using random forests in addition to the main model serves several key purposes:

- **Validation and Robustness**: The random forest model validates findings from the regression model by identifying the most influential predictors of stop-and-search activity without assuming specific linear relationships. If similar predictors emerge as significant in both models, this strengthens the evidence for their role in influencing stop-and-search practices.
- **Non-linear and Interaction Effects**: Random forests are effective at capturing non-linear relationships and variable interactions, allowing for deeper insights into how combinations of factors (e.g., ethnicity and region) may contribute to disparities in stop-and-search practices. This is particularly relevant for understanding complex social phenomena where the effects of variables are unlikely to be purely linear or independent.
- **Variable Importance**: Unlike regression coefficients, which quantify the effect of each predictor, random forests provide variable importance scores. These scores allow us to rank predictors by their influence on the outcome (total stop-and-search counts), offering an alternative perspective on which factors most drive stop-and-search activity.

By using random forests alongside the main regression model, we obtain a comprehensive view of stop-and-search patterns, combining the interpretability of regression with the flexibility and interaction-capturing ability of random forests.

4.7.2 Overview of the Random Forest Model

The random forest model applied in this study is designed to predict **total stop-and-search counts** using demographic, geographic, and contextual variables. Random forests operate by constructing an ensemble of decision trees, each built from a bootstrap sample of the original data. Each tree independently predicts the outcome, and the final prediction for each observation is obtained by aggregating the predictions across all trees (for regression, this is typically the mean prediction). Key steps and features of the random forest model are as follows:

1. **Bootstrap Sampling and Tree Construction**:
    - For each decision tree in the forest, a random sample (with replacement) is drawn from the dataset, which is known as bootstrap sampling. This approach introduces variation among the trees, enhancing the model's robustness and reducing overfitting to any one sample.

- At each node of a tree, a random subset of predictors is selected as candidate variables for splitting. This process, known as **feature bagging**, ensures that each tree explores different predictor combinations, further improving the model's generalization to new data.

2. **Split Criteria and Aggregation**:
    - The model uses **mean squared error (MSE)** as the criterion for creating splits in each tree, which minimizes the variance within each split group. This criterion is effective for identifying variable thresholds associated with differences in stop-and-search counts.
    - Once all trees are built, the model aggregates predictions from each tree to produce a single, overall prediction for each observation. This aggregation (averaging) process improves accuracy and stability, as it reduces the likelihood of overfitting that may occur in individual decision trees.
3. **Variable Importance Measures**:
    - The random forest model provides **variable importance** scores, specifically % Increase in Mean Squared Error (%IncMSE), which indicates the decrease in prediction accuracy when a variable is removed. Higher %IncMSE values suggest that the variable is crucial for accurate prediction, while lower or negative values indicate limited or no contribution to the model's predictive ability.
    - These importance scores allow us to rank the predictors, offering insights into which factors most strongly influence stop-and-search counts. The importance measures are central to our objectives, as they reveal key drivers of disparities and patterns in stop-and-search practices across different groups.

4.7.3 Interpretation of Random Forest Output

In the results, we focus on the variable importance scores to identify the most influential factors contributing to stop-and-search activity. This approach allows us to highlight demographic, regional, and contextual factors that may not show straightforward relationships but still play a significant role in total search volume. The random forest findings are interpreted in conjunction with the main regression model results to provide a comprehensive analysis of stop-and-search practices, validating key factors and uncovering complex relationships that inform policy recommendations.

By incorporating the random forest model into the analysis, we achieve a richer, multi-faceted understanding of stop-and-search dynamics, capturing both straightforward and complex patterns that contribute to observed disparities. This dual-model approach thus enhances the robustness and depth of the study's findings, aligning with the project's objective of identifying and addressing factors contributing to inequality in policing practices.

5 Results

The results presented in this section provide a comprehensive analysis of stop-and-search data in England and Wales. Leveraging both descriptive statistics and advanced statistical models, we explore the key patterns, disparities, and trends in stop-and-search practices across demographic, geographic, and temporal dimensions. The findings highlight significant disproportionalities in stop-and-search rates among ethnic groups, with a particular focus on the Black or Black British population. Additionally, the results provide insights into the influence of factors such as age, sex, region, and reason for search on stop-and-search frequencies. This section concludes with the application of machine learning techniques, specifically random forests, to uncover additional patterns and establish a framework for prediction.

5.1 Census data

The analysis of census data provides critical insights into the demographic composition of England and Wales. This section examines the distribution of ethnic groups across English regions, highlighting variations in population proportions and regional disparities. The results form the basis for understanding broader demographic patterns that intersect with stop-and-search practices. Key metrics such as mean percentages, standard deviations, and ranges are calculated to quantify these distributions and inform subsequent analyses.

5.1.1 Ethnic Composition Across English Regions

The table below summarizes the composition of ethnic groups across English regions, showing the mean percentage, standard deviation (SD), and range (minimum to maximum) for each group. This provides an overview of the variability in ethnic group proportions across regions.

	Ethnic Group Composition Across English Regions (%)			
Ethnic Group	mean	sd	min	max
Asian or Asian British	8.2	5.4	2.8	20.7
Black or Black British	3.4	3.7	0.9	13.5
Mixed	2.6	1.2	1.3	5.7
Other Ethnic Group	1.8	1.6	0.9	6.3
White	84.0	11.8	53.8	93.8

The data highlights significant disparities between ethnic groups. Notably, the average proportion of individuals identifying as Black or Black British is 3.4%, with regional variation ranging from 0.9% to 13.5%. In contrast, the White group, which has the largest mean proportion at 84%, also shows variability, ranging from 53.8% to 93.8%. These patterns underline the diverse ethnic composition across regions, particularly the stark contrast between Black and White populations.

5.1.2 Population Distribution by Ethnic Group and Region

The table below summarizes the total population distribution by ethnic group across regions in England and Wales, based on census data. Population counts are presented in thousands, providing a clear view of regional demographics. The summary includes total counts for each ethnic group and highlights regional variations, which are key to understanding the geographic diversity of the population.

	Total Population by Ethnic Group and Region (per 1,000)				
region	Asian or Asian British	Black or Black British	Mixed	Other Ethnic Group	White
East Midlands	391.1	130.0	117.2	61.9	4179.8
Eastern	405.9	185.0	179.7	86.2	5478.4
London	1817.6	1188.4	505.8	556.8	4731.2
North East	98.0	26.6	33.3	26.3	2462.7
North West	622.7	173.9	163.3	110.2	6347.4
South East	650.5	221.6	260.9	135.7	8009.4
South West	159.2	69.6	114.1	48.7	5309.6
Wales	89.0	27.6	48.6	26.5	2915.9
West Midlands	794.3	269.0	178.2	124.2	4585.0
Yorkshire and the Humber	487.1	117.6	117.0	79.1	4680.0
Total	**5515.4**	**2409.3**	**1718.1**	**1255.6**	**48699.4**

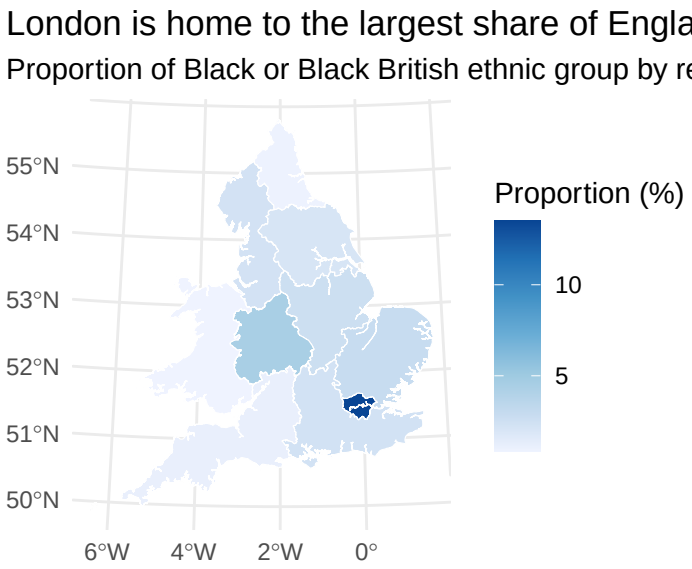
The census data reveals that the total population of individuals identifying as Black or Black British across all regions is 2,409 thousand, compared to 48,699 thousand for the White group. In London, which has the highest concentration of Black or Black British individuals among all regions, this group accounts for 1,188 thousand, while the White population in London is 4,731 thousand.

The region with the largest White population is South East, while London has the highest Black or Black British population outside London. These figures highlight the substantial geographic and demographic diversity captured in the census data.

5.1.3 Geographic Distribution of the Black or Black British Population

The map below visualizes the geographic distribution of individuals identifying as Black or Black British across regions in the UK, based on census data. It represents the proportion

of this ethnic group within each region as a percentage of the total regional population, highlighting spatial patterns and regional disparities.



; 2021, licensed under the Open Government Licence v3.0.
Statistics data © Crown copyright and database right 2026.
ıtains OS data © Crown copyright and database right 2026.

The map illustrates substantial regional variability in the proportion of individuals identifying as Black or Black British. The region with the highest proportion is London, where this group accounts for 13.5% of the total population. In contrast, the region with the lowest proportion is Wales, at just 0.9%.

These results emphasize the uneven distribution of the Black population across the UK, reflecting broader patterns of settlement, migration, and urbanization.

5.2 Stop and Search data

This section examines police stop and search activities across England and Wales. This data includes searches conducted under key legislations, specifically Section 1 of the Police and Criminal Evidence Act (PACE) and Section 60 of the Criminal Justice and Public Order Act (CJPOA). By analyzing annual trends, legislative frameworks, and demographic characteristics, this section aims to provide insights into the scale and patterns of stop and search practices. The results inform subsequent analyses of disparities and contributing factors.

5.2.1 Total Stop and Searches by Year

The table below details the annual total of stop and searches conducted in England and Wales under Section 1 of the Police and Criminal Evidence Act (PACE) and Section 60 of

the Criminal Justice and Public Order Act (CJPOA). This summary highlights year-on-year trends and the overall scale of these police powers.

Total Stop and Searches Conducted in England and Wales by Financial Year	
Financial Year	Total Searches
2020/21	714,914
2021/22	530,970
2022/23	547,000
2023/24	536,151
2024/25	528,242

The data shows that the highest number of stop and searches was conducted in the financial year 2020/21, with a total of 714,914 searches. In contrast, the financial year with the lowest number of searches was 2024/25, with 528,242 searches.

These figures reflect changes in policing strategies and legal contexts, as well as broader socio-political factors influencing the use of stop and search powers over time.

i Analysis sample vs. published totals

The totals above match the figures published by the UK Government and include *all* recorded stop and searches under PACE s.1 and CJPOA s.60. The disparity analysis presented in the rest of this report uses a smaller **analysis sample** that excludes three categories of records that cannot be assigned to a demographic group or geographic denominator (see Section 4.3 for details):

1. **Vehicle searches**: no person-level demographics are recorded.
2. **British Transport Police (region 0)**: no Census population denominator exists for BTP.
3. **"Not Stated" ethnicity**:these records cannot be attributed to any ethnic group for rate calculations.

Reconciliation of Published Totals and Analysis Sample				
Financial Year	Published Total	Analysis Sample	Excluded	% Retained
2020/21	714,914	555,915	158,999	77.8
2021/22	530,970	405,946	125,024	76.5
2022/23	547,000	411,353	135,647	75.2
2023/24	536,151	404,636	131,515	75.5
2024/25	528,242	402,376	125,866	76.2

The analysis sample retains approximately 76% of all searches across years. All disparity rates, IRRs, and dashboard figures are based on this analysis sample.

Note that the stop and search dataset contains some missing data, primarily in the `officer_defined_ethnicity` variable. Other missing values occur because certain searches were related to vehicles, which are not included in this analysis. As a result, the percentage

of truly missing data in the original dataset is 4.6%. These missing values were handled appropriately to ensure robust analysis.

5.2.2 Disparity Rates Relative to White Individuals

The table below presents disparity rates for stop and searches relative to White individuals across different financial years. Disparity rates indicate how much more (or less) likely individuals from other ethnic groups are to be stopped and searched compared to White individuals. These rates are derived by comparing the stop-and-search rate per population for each ethnic group against the rate for White individuals.

	Disparity rates relative to White individuals, by financial year			
Financial Year	Asian or Asian British	Black or Black British	Mixed	Other Ethnic Group
2020/21	1.85	5.49	1.67	1.24
2021/22	1.59	4.81	1.67	1.21
2022/23	1.43	4.08	1.64	1.26
2023/24	1.34	3.62	1.62	1.35
2024/25	1.39	3.65	1.74	1.47
Average	**1.52**	**4.33**	**1.67**	**1.31**

The data reveals that individuals identifying as Black or Black British consistently experience the highest disparity rates in stop and searches compared to White individuals. The highest disparity was recorded in 2020/21, with a relative rate of 5.49, while the lowest disparity occurred in 2023/24, at 3.62. On average, the disparity rate for this group across all financial years is 4.33.

These figures underscore the significant disparities faced by Black individuals in stop and search practices. Similar trends are observed for other ethnic groups, though with lower disparity rates compared to the Black or Black British group.

The most recent year, 2024/25, yields a Black/White disparity rate of **3.6**× (from 3.65). This is the unadjusted figure, calculated directly from search counts divided by Census 2021 population denominators with no covariates controlled, and is the number shown on the interactive dashboard. The adjusted incidence rate ratio from the Poisson model (see Section 5.6) is lower because the model compares ethnic groups within the same region, and Black residents are concentrated in London, where search rates are high for all groups; the gap between the two measures reflects that regional composition rather than any inconsistency in the data.

5.2.3 Outcome and Link

The question we want to investigate here is whether stop and search results in successful motives. We have removed “Unknown link” and “Unknown whether article found.” Then,

we create a new column called “success” - if link is “Linked,” it is successful, otherwise, it is not.

The overall success ratio is: 34.5%.

5.3 Modelling

This section outlines the statistical modelling approach applied to analyze stop and search data. Multiple models were constructed and evaluated, each targeting specific aspects of the data while considering methodological rigor. The focus was on understanding disparities across financial years, regions, and ethnic groups, including key covariates such as sex, age group, and reason for search.

5.3.1 Overview of Models Tested

1. **Model 1: Simple Linear Regression**
    - A baseline linear regression model was fitted with the stop-and-search rate (`rate`) as the dependent variable.
    - Independent variables included financial year, region, ethnicity, sex, age group, and reason for search.
    - Diagnostics revealed significant heteroscedasticity, making this model unsuitable.
2. **Model 2: Log-Transformed Linear Regression**
    - To address heteroscedasticity, a logarithmic transformation of the rate (`log(rate)`) was applied.
    - The transformed model demonstrated improved fit and approximate normality in residuals.
3. **Mixed Linear Model**
    - A mixed-effects model was considered, treating financial year as a random effect to account for potential year-specific variability.
    - However, the random effect’s variance was close to zero, indicating that this added complexity was unnecessary.
4. **Generalized Linear Model (GLM) with Offset**
    - A Poisson GLM with a log link function was employed, using the number of searches as the dependent variable and the log population as an offset.
    - This model was particularly relevant for adjusting for population size across regions and ethnic groups.
    - All parameters were highly significant (`p < 0.0001`), and the effect sizes for ethnic groups were consistent with findings from prior research.
5. **Model Evaluation**

- Pseudo-R^2 by Cohen was used to assess the GLM's fit. The pseudo-R^2 value confirmed the model's adequacy for explaining variability in the data.

5.3.2 Key Considerations

- **Model Selection**: While multiple models were tested, the GLM with an offset was identified as the most appropriate for the dataset, balancing interpretability and statistical validity.
- **Diagnostics**: Each model was assessed for assumptions such as normality of residuals, heteroscedasticity, and random effects variance to ensure robustness.
- **Significance of Covariates**: Across models, ethnicity consistently emerged as a significant factor, supporting the focus on disparities in stop-and-search practices.

These models provide a robust framework for understanding patterns and disparities in stop-and-search data, and they form the foundation for further statistical and interpretive analysis.

Based on the results of the evaluated models, a Poisson regression with a population offset was selected as the final model. The following section delves into the rationale, structure, and key findings of the chosen model.

5.4 Chosen Model

5.4.1 Overview of the Poisson Regression

The Poisson regression model with an offset for population size was selected as the most appropriate approach for this analysis. This model accounts for varying population sizes across regions and ethnic groups, allowing for a fair comparison of stop-and-search rates. The dependent variable is the total number of stop and searches, with independent variables including financial year, region, ethnicity, sex, age group, and reason for search. Reference levels for categorical variables are:

- Financial Year: 2020/21
- Region: East Midlands
- Ethnicity: White
- Sex: Female
- Age Group: 25 and above
- Reason for Search: Non-drug-related

5.4.2 Model results

The table below summarizes the results of the Poisson regression model analyzing the frequency of stop and searches across England and Wales. The raw coefficients from the model (log-scale) provide statistical estimates for the association between each variable and the expected log count of stop and searches.

However, for ease of interpretation, these coefficients are exponentiated to present rate ratios, which quantify the multiplicative effect of each variable relative to the baseline categories. For instance, a rate ratio greater than 1 indicates a higher likelihood of stop and searches compared to the baseline category, while a rate ratio less than 1 indicates a lower likelihood. Baseline levels for the categorical variables are: financial year **2020/21**, region **East Midlands**, ethnicity **White**, sex **Female**, age group **25 and above**, and reason for search **non-drug-related**.

The following tables first present the raw coefficients for statistical rigor, followed by the exponentiated coefficients (rate ratios) for practical interpretation.

5.4.2.1 Raw Coefficients

Table 5.6: Raw Coefficients from Quasi-Poisson model (selected terms)

Term	Log Coef.	Std Error	Z	P-Value	Lower CI	Upper CI
financial_year2021/22	-0.319	0.0282	-11.31	0.000	-0.375	-0.264
financial_year2022/23	-0.307	0.0281	-10.92	0.000	-0.362	-0.252
financial_year2023/24	-0.332	0.0283	-11.72	0.000	-0.388	-0.277
financial_year2024/25	-0.365	0.0286	-12.74	0.000	-0.421	-0.309
ethnicityAsian	0.156	0.0306	5.11	0.000	0.096	0.216
ethnicityBlack	1.106	0.0291	38.07	0.000	1.049	1.163
ethnicityMixed	0.306	0.0488	6.26	0.000	0.209	0.400
ethnicityOther	-0.069	0.0635	-1.08	0.280	-0.195	0.054
sexMale	2.080	0.0296	70.19	0.000	2.022	2.138
age_under25Under 25	0.025	0.0186	1.36	0.174	-0.011	0.062
reason_drugsDrugs	0.573	0.0194	29.58	0.000	0.535	0.611

5.4.2.2 Incidence Rate Ratios

Table 5.7: Key Incidence Rate Ratios from Quasi-Poisson model (selected terms)

Term	IRR	Lower CI	Upper CI
financial_year2021/22	0.73	0.69	0.77
financial_year2022/23	0.74	0.70	0.78
financial_year2023/24	0.72	0.68	0.76
financial_year2024/25	0.69	0.66	0.73
ethnicityAsian	1.17	1.10	1.24
ethnicityBlack	3.02	2.85	3.20

ethnicityMixed	1.36	1.23	1.49
ethnicityOther	0.93	0.82	1.06
sexMale	8.00	7.55	8.49
age_under25Under 25	1.03	0.99	1.06
reason_drugsDrugs	1.77	1.71	1.84

5.4.2.3 Key Findings

The results show significant disparities in stop-and-search likelihoods. For instance, individuals identifying as **Black or Black British** are 3.02 times more likely to be stopped and searched compared to **White** individuals, with a 95% confidence interval of (2.85, 3.2). Similarly, **males** are 8 times more likely to be stopped and searched than females, highlighting a significant gender disparity.

London stands out with the highest regional rate: individuals there are 4.09 times as likely to be stopped and searched as those in the East Midlands (95% CI: 3.7–4.54).

These findings underscore critical disparities in stop-and-search practices across demographic and geographic dimensions.

5.4.3 Model Coefficients: Interpretation

This analysis evaluates the association between demographic, temporal, and geographic factors and police stop-and-search frequencies in England and Wales using a Poisson regression model. The model includes an offset for population size to account for differences in regional populations, providing rate ratios for stop-and-search frequencies across groups.

Key variables were encoded for interpretability: **age group** was binary, with "Under 25" for individuals aged 10–24 and "25 and over" for those 25 and above; **reason for search** was binary, with "Drugs" for drug-related stops and "Non-drugs" for other reasons. Baseline levels for categorical variables include: financial year **2020/21**, region **East Midlands**, ethnicity **White**, sex **Female**, age group **25 and above**, and reason for search **non-drug-related**. All coefficients are exponentiated to represent rate ratios.

Each rate ratio indicates the multiplicative effect of a variable on the likelihood of stop and search, relative to the baseline. Values greater than 1 represent increased likelihood, while values below 1 signify decreased likelihood.

5.4.3.1 Coefficients and Interpretations

5.4.3.1.1 Financial Year

- **2021/22**: The rate ratio for **2021/22** is 0.73 (95% CI: 0.69–0.77), indicating that stop-and-search frequencies were approximately 0.73 times as likely compared to **2020/21**.
- **2022/23**: The rate ratio for **2022/23** is 0.74 (95% CI: 0.7–0.78).

These results suggest a consistent decrease in stop-and-search activities over time.

5.4.3.1.2 Region (Relative to East Midlands)

Most regions show higher stop-and-search rates than East Midlands, with **London** standing out. For instance: - **London**: Rate ratio of 4.09 (95% CI: 3.7–4.54). - **South East**: Rate ratio of 1.44 (95% CI: 1.28–1.61).

The variation across regions likely reflects differences in policing practices, crime profiles, and population density.

5.4.3.1.3 Ethnicity (Relative to White)

Ethnic minorities showed significantly different rates of stop and search compared to White individuals: - **Black or Black British**: Rate ratio of 3.02 (95% CI: 2.85–3.2), indicating a 202.2% higher likelihood of being stopped and searched.

5.4.3.1.4 Sex (Relative to Female)

- **Male**: Rate ratio of 8 (95% CI: 7.55–8.49). Men are significantly more likely to be stopped and searched compared to women.

5.4.3.1.5 Age Group (Relative to Age 25+)

- **Under 25**: Rate ratio of 1.03 (95% CI: 0.99–1.06), reflecting a slightly increased likelihood for younger individuals.

5.4.3.1.6 Reason for Search (Relative to Non-Drug Related)

- **Drug-Related**: Rate ratio of 1.77 (95% CI: 1.71–1.84), indicating a substantial increase in stop-and-search likelihood for drug-related reasons.

5.4.4 Model Fit and Diagnostics

- **Pseudo R-squared (Cohen's)**: The pseudo R-squared value of 0.884 indicates that the model explains approximately 88.4% of the variability in stop and search counts, suggesting a strong fit. This high explanatory power supports the model's validity in identifying and quantifying the associations between stop and search practices and demographic, geographic, and temporal factors.

The residual deviance (342,749.3 on 1950 degrees of freedom) indicates a good model fit, with some remaining variability potentially attributable to unobserved factors. The estimated dispersion parameter is 181.8, confirming severe overdispersion; standard errors are scaled accordingly by the Quasi-Poisson family (see Methodology). AIC is not defined for Quasi-Poisson models.

- **Interaction model (ethnicity × financial year)**: To account for part of this overdispersion and allow ethnic disparities to vary over time, we fitted an extended model including an `ethnicity × financial_year` interaction. The interaction model reduces the dispersion parameter from 181.8 to 174.9 and improves the pseudo R^2 from 88.4% to 88.8%. An F-test confirms the interaction terms are jointly significant ($p < 0.001$), indicating that ethnic disparities have shifted meaningfully across financial years. Despite this improvement, the remaining overdispersion ($\hat{\phi} \approx 175$) reflects heterogeneity at the force level and other unmodelled factors, a common feature of aggregate policing data. The Quasi-Poisson framework ensures that all reported confidence intervals and significance tests remain conservative regardless.

5.4.5 Addressing the Project Questions

1. **Do Black People in Metropolitan Areas Experience More Drug Stop and Searches?**

   To address this question, we analyze the combined effects of ethnicity (Black or Black British), region (London as a proxy for metropolitan area), and reason for search (drug-related):

   - **Ethnicity (Black or Black British)**: The rate ratio is 3.02 (95% CI: 2.85–3.2), indicating Black individuals are approximately 3.02 times more likely to be stopped and searched compared to White individuals.
   - **Region (London)**: The rate ratio for London relative to East Midlands is 4.09 (95% CI: 3.7–4.54), confirming that London has far higher search rates than the baseline region.
   - **Reason for Search (Drug-Related)**: The rate ratio is 1.77 (95% CI: 1.71–1.84), indicating drug-related searches are approximately 1.77 times more likely than non-drug-related searches.

   Combined effect: The multiplicative rate effect for Black individuals in London during drug-related stops is approximately 21.95 times that for White individuals in the East Midlands for non-drug-related reasons, holding sex and age constant.

2. **What is the Relationship Between Young People Under 25 and Drug Stops?**

   For this question, we examine the effects of **age group (under 25)** and **reason for search (drug-related)**:

   - **Age Group (Under 25)**: The rate ratio is 1.03 (95% CI: 0.99–1.06), indicating individuals under 25 are approximately 1.03 times more likely to be stopped and searched compared to individuals aged 25 and above.
   - **Reason for Search (Drug-Related)**: The rate ratio is 1.77 (95% CI: 1.71–1.84), suggesting that drug-related searches are approximately 1.77 times more likely than non-drug-related searches.

Combined effect: The multiplicative rate effect for individuals under 25 in drug-related stops is approximately 1.82 times more likely than for older individuals for non-drug-related reasons.

5.4.6 Implications of Findings

This analysis addresses two primary questions regarding drug-related stop-and-search practices in metropolitan areas:

1. **Do Black Males under 25 Years of Age in Metropolitan Areas Experience More Drug Stop-and-Searches?**
2. **How Does This Compare to Black Females, White Females, and White Males under 25 in Metropolitan Areas, Specifically When the Reason for Stop and Search is Drug-Related?**

5.4.6.1 1. Black Males Under 25 in Metropolitan Areas

The model results indicate that **Black or Black British males under 25 years of age** in metropolitan areas are significantly more likely to experience drug-related stop-and-search encounters. Specifically, this group serves as the reference point for comparison against other demographic groups.

5.4.6.2 2. Comparative Analysis Across Demographic Groups

The model's statistical reference cell is a White female aged 25+ stopped for non-drug reasons in the East Midlands (i.e. the intercept). While useful for model construction, the most **policy-relevant** comparison is between ethnic groups within the same sex, age group, and search reason. Because the population offset is by region and ethnic group, this compares searches per resident of each ethnic group in the region, not per resident of that sex and age group (Section B.4 gives the comparison with finer denominators). The table below re-expresses the combined rate ratios in two ways: first relative to **White individuals of the same sex, age group, and reason**, then relative to the statistical reference cell.

Demographic Profile	Black/White Ratio (same sex, age, reason)	vs Reference Cell (White Female, 25+,
Black Male Under 25, Drug Stop	**3.02×**	**44.01×**
Black Male 25+, Drug Stop	3.02×	42.91×
White Male Under 25, Drug Stop	—	14.56×
White Male 25+, Drug Stop	—	14.2×
Black Female 25+, Drug Stop	3.02×	5.36×
White Female 25+, Drug Stop	—	1.77×

Interpretation:

- Because this is a main-effects (additive on the log scale) model, the **ethnicity multiplier is constant**: Black individuals are searched 3.02× as often per resident as White individuals in the same region, for every combination of sex, age group, and search reason. This 3.02× factor is the central measure of racial disparity.
- A **Black male under 25** stopped for drug-related reasons has a combined rate ratio of **44.01** relative to the reference cell, but **3.02**× relative to a White male under 25 stopped for the same reason. The difference between 44.01 and 3.02 is entirely explained by the sex (8×), age (1.03×), and drug (1.77×) effects, which affect Black and White individuals equally in this model.
- **White Males Under 25 for Drugs** have a rate ratio of 14.56 versus the reference cell, highlighting that sex and drug effects are substantial independent of ethnicity, but the **ethnicity gap** roughly triples the rate for Black individuals in every combination.

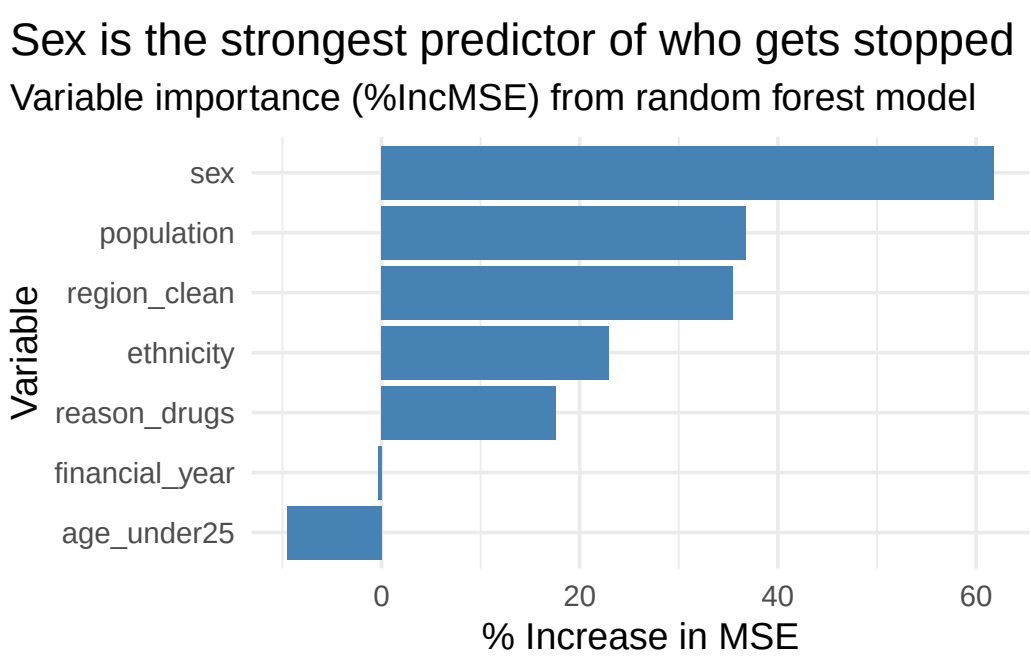
5.4.7 Policy Implications

The identified disparities necessitate targeted interventions to ensure equitable policing practices:

1. **Policy Review and Reform:**
    - **Implement Equitable Policing Guidelines:** Develop and enforce guidelines that minimize racial and gender biases in stop-and-search operations.
    - **Data Transparency:** Mandate comprehensive data collection and public reporting on stop-and-search encounters to monitor and address disparities.
2. **Training and Awareness:**
    - **Bias Training:** Enhance training programs for law enforcement to recognize and mitigate unconscious biases that contribute to disproportionate targeting of specific demographic groups.
    - **Cultural Competency:** Foster cultural competency among police officers to improve interactions with diverse communities.
3. **Community Engagement:**
    - **Build Trust:** Engage with communities, particularly among young Black males, to build trust and cooperation between law enforcement and the populations they serve.
    - **Feedback Mechanisms:** Establish channels for community members to provide feedback and report instances of discriminatory practices without fear of retaliation.

By addressing these areas, policymakers and law enforcement agencies can work towards more fair and just policing strategies, ensuring that stop-and-search practices do not disproportionately affect specific demographic groups and that all individuals are treated equitably under the law.

5.5 Random Forest Model: Machine Learning Approach



A random forest model was used to identify the most influential factors associated with total stop-and-search activity across financial years, regions, and demographic groups. This approach complements the regression analysis by revealing complex, non-linear relationships between predictors and total searches. The random forest was trained on demographic and contextual variables, including **financial year**, **region**, **ethnicity**, **sex**, **age group**, **reason for search**, and **population size**, to determine the relative importance of each factor in predicting the total number of stop-and-searches. The variable importance measures (based on %IncMSE) highlight key factors that influence stop-and-search volume and allow an in-depth exploration of the project's objectives.

5.5.1 Variable Importance and Interpretation

The variable importance plot shows each predictor's contribution to the model, measured by % Increase in Mean Squared Error (%IncMSE). Key findings from the random forest analysis include:

1. **Sex**:
    - **Sex** emerged as the most influential predictor (%IncMSE = 61.8), the single strongest driver of search volume in the model. Consistent with the regression findings, males are disproportionately represented in stop-and-search activity. This alignment across model types underscores the persistent gender gap in policing.
2. **Population**:

- **Population size** was the second most important predictor (%IncMSE = 36.8). Areas with larger populations experience higher total search counts, reflecting the concentration of searches in densely populated regions.

3. **Region**:
    - **Region** also displayed a high %IncMSE, suggesting that geographic location strongly influences stop-and-search activity. This regional variation likely reflects different policing strategies and priorities across England and Wales. The substantial impact of region on search volume suggests that certain areas are more intensely policed, which may be related to local crime rates, population density, or policy differences.
4. **Ethnicity**:
    - **Ethnicity** emerged as an influential factor, with notable %IncMSE values. Within the random forest model, Black or Black British individuals were particularly associated with higher search counts, reflecting the model's capacity to capture ethnic disparities in policing. The importance of ethnicity aligns with the project's focus on identifying demographic disparities in stop-and-search practices, confirming that specific ethnic groups are more frequently targeted in police searches.
5. **Financial Year and Reason for Search**:
    - **Financial year** and **reason for search** also contributed to the prediction of total searches, though to a lesser extent than population, sex, and region. The financial year variable's influence suggests temporal changes in policing activity, potentially linked to policy adjustments or shifts in crime prevention focus over time. **Drug-related stops** (indicated by the reason for search variable) were also predictive of higher search counts, suggesting that specific search reasons contribute to variation in stop-and-search volume.
6. **Age Group**:
    - **Age group** exhibited a low and even negative %IncMSE, indicating that age group alone may not meaningfully enhance prediction accuracy in this model. This contrasts with the regression analysis, where age group had a positive association with search counts. The random forest results suggest that age might interact with other factors (like ethnicity or reason for search) rather than being a primary predictor of search volume independently.

5.5.2 Implications for Project Objectives

The random forest model findings align with the project's objectives, providing detailed insights into factors influencing total stop-and-search counts and validating key areas of demographic disparity. The high importance of **sex**, **population**, and **region** reveals structural factors that drive search volumes across England and Wales, with clear implications for regional and demographic disparities in policing. The consistent influence of **ethnicity** across both the random forest and regression models underscores the model's reliability in

identifying areas where certain groups are disproportionately affected by stop-and-search practices.

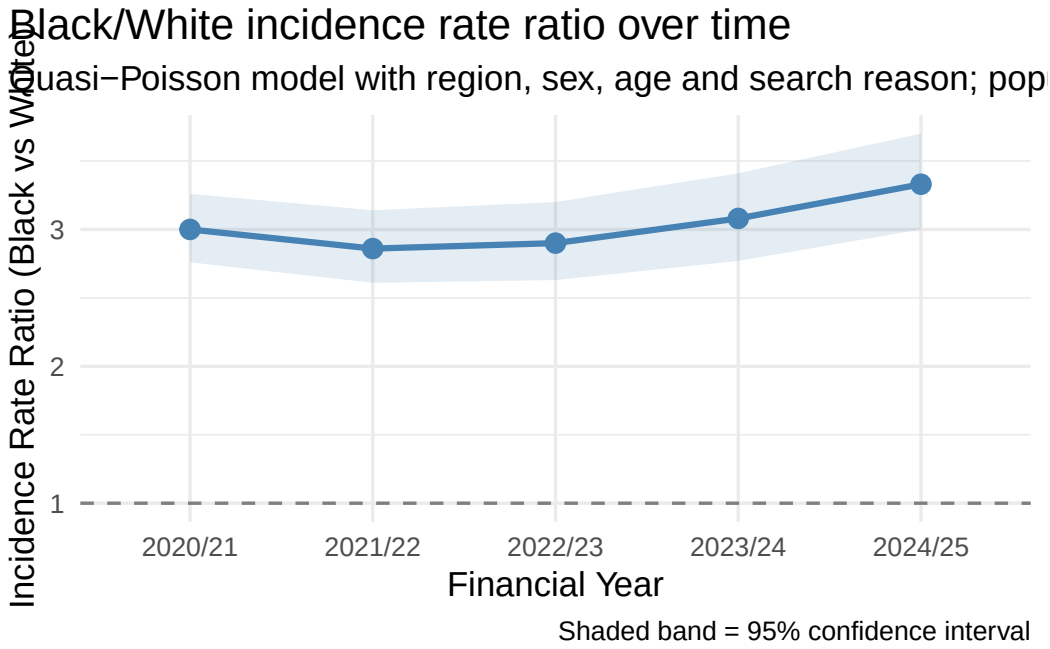
In summary, the random forest analysis provides a robust exploration of the factors driving total stop-and-search volume. By identifying sex, population, region, and ethnicity as major predictors, the model highlights demographic and regional differences in policing practices. This evidence-based approach supports the project's goal to uncover disparities and guide recommendations for more equitable policing policies across demographic groups.

5.6 Five-Year Disparity Trend

The sections above present the pooled model results. Here we examine how the **adjusted disparity**, from a model that includes region, sex, age band and reason for search, with population by region and ethnic group as the offset, has evolved year by year over the five-year period 2020/21–2024/25, using pre-computed per-year Quasi-Poisson models.

Table 5.9: Per-year adjusted IRR for Black vs White individuals (Quasi-Poisson)

Financial Year	Adjusted IRR	95% CI
2020/21	3.00	(2.76–3.26)
2021/22	2.86	(2.61–3.14)
2022/23	2.90	(2.63–3.2)
2023/24	3.08	(2.77–3.41)
2024/25	3.33	(3–3.7)



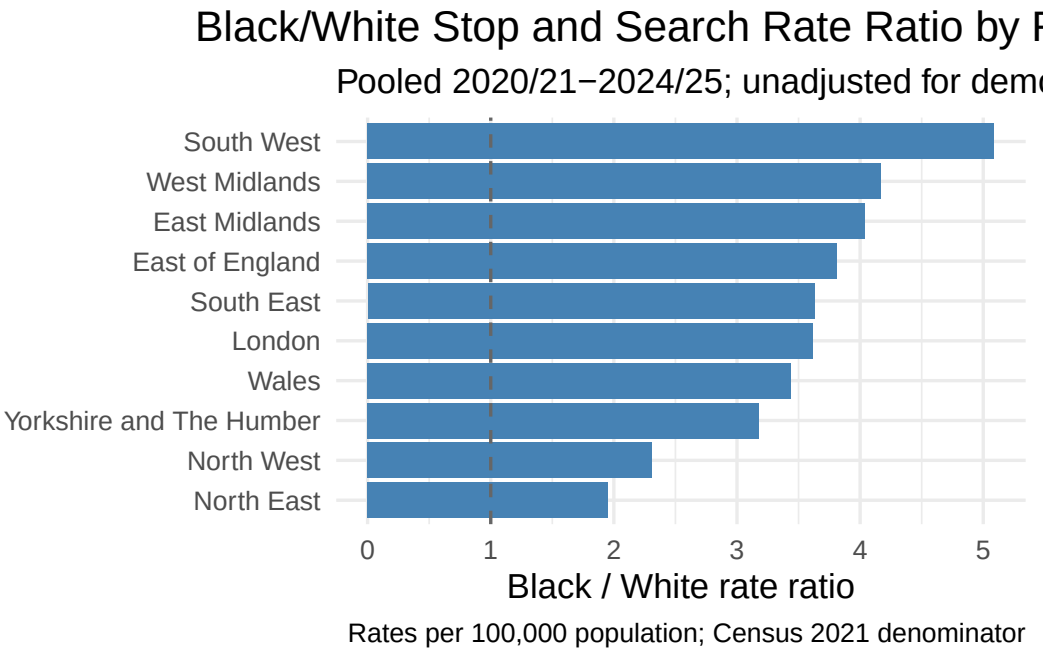
The adjusted IRR for Black individuals relative to White individuals shows a **U-shaped trend**: falling slightly in 2021/22 (IRR = 2.86) and 2022/23 (IRR = 2.9) before rising sharply to 3.08 in 2023/24 and reaching 3.33 (95% CI: 3–3.7) in 2024/25. This compares with 3 (95% CI: 2.76–3.26) in 2020/21 (p = 0.13 for the difference). With population also split by sex and age band (Section B.4), the corresponding estimates are 2.67 and 3.05 (p = 0.049).

These adjusted ratios sit against a backdrop of **falling overall search volumes**. Raw Black search rates per 1,000 population fell faster than White rates between 2020/21 and 2024/25, which is why the raw ratio narrowed. Most of that fall was in London, where almost half of Black residents live; within London, and across the rest of England and Wales, Black searches fell less than White searches or rose, so the adjusted ratio, which compares groups within regions, did not narrow.

5.7 Regional Force Rankings

Stop and search rates and Black/White disparity ratios by region (2020/21–2024/25 pooled)

Region	Black rate (per 100k)	White rate (per 100k)	Black/White ratio
South West	**12.2**	**2.4**	**5.09**
West Midlands	15.4	3.7	4.17
East Midlands	10.9	2.7	4.04
East of England	12.9	3.4	3.81
South East	16.3	4.5	3.63
London	24.7	6.8	3.62
Wales	13.6	4.0	3.44
Yorkshire and The Humber	14.3	4.5	3.18
North West	37.6	16.3	2.31
North East	12.9	6.6	1.95



The regional analysis reveals striking variation in the unadjusted Black/White disparity. The **South West** has the highest ratio at 5.09×, substantially exceeding London (3.62×). This is notable given that London is typically assumed to be the focal point of ethnic disproportionality. The North East has the lowest ratio (1.95×), though it remains above parity. The South West anomaly is discussed further in the Discussion chapter.

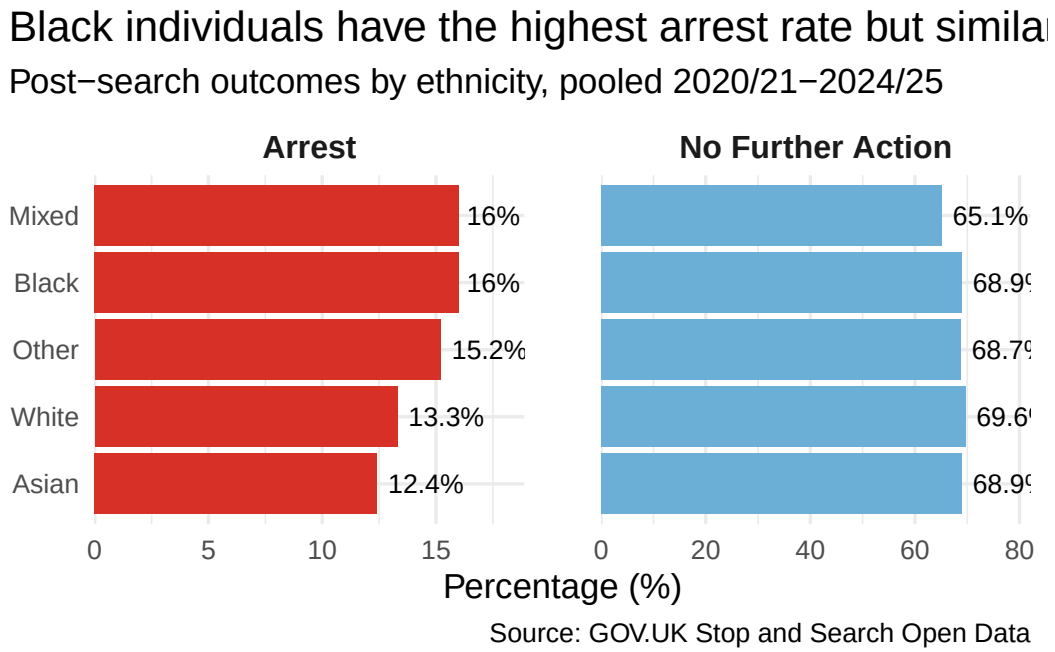
The North West shows extremely high absolute rates for all ethnic groups (Black: 37.6 per 100,000; White: 16.3 per 100,000), suggesting a general pattern of intensive policing rather than ethnicity-specific targeting, reflected in its relatively lower ratio of 2.31×.

5.8 Outcome Disparity by Ethnicity

A key indicator of whether stop and search powers are deployed appropriately is the **post-search outcome distribution**. If the evidential threshold for initiating a search is genuinely equivalent across ethnic groups, we would expect broadly similar outcome profiles.

Post-search outcome distribution by ethnicity (2020/21–2024/25, %, selected outcomes)

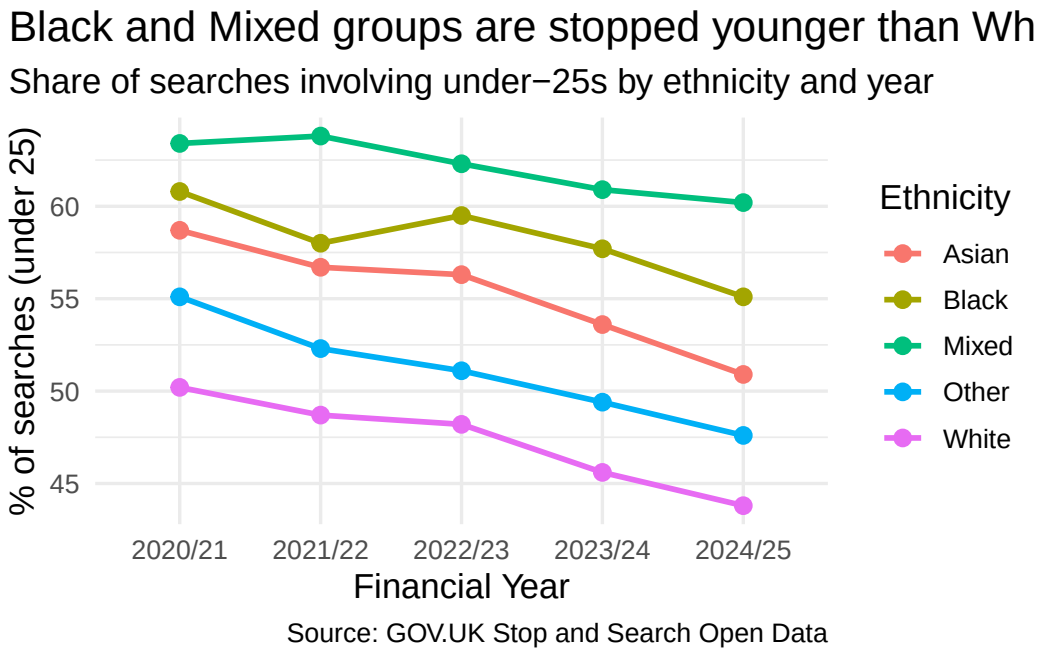
ethnicity	Arrest (%)	Community Resolution (%)	No Further Action (%)
Asian	12.4	9.5	68.9
Black	16.0	8.7	68.9
Mixed	16.0	10.6	65.1
Other	15.2	9.6	68.7
White	13.3	7.9	69.6



Across all ethnic groups, **No Further Action (NFA)** is by far the most common outcome, indicating that the majority of searches do not result in any enforcement action. The NFA rate for Black individuals (68.9%) is marginally lower than for White individuals (69.6%), which might appear reassuring. However, the **arrest rate** tells a different story: Black individuals have one of the highest arrest rates at 16% (tied with Mixed), compared to 13.3% for White individuals and 12.4% for Asian individuals. This differential is consistent with searches against Black individuals being somewhat more likely to yield enforcement action; yet given the substantially higher stop rate, it also implies that a large number of Black individuals are searched without justification. The implications of this pattern are discussed in the Discussion chapter.

5.9 Age by Ethnicity Interaction

Percentage of stops involving individuals under 25, by ethnicity and year (%)					
Financial Year	Asian	Black	Mixed	Other	White
2020/21	58.7	60.8	63.4	55.1	50.2
2021/22	56.7	58.0	63.8	52.3	48.7
2022/23	56.3	59.5	62.3	51.1	48.2
2023/24	53.6	57.7	60.9	49.4	45.6
2024/25	50.9	55.1	60.2	47.6	43.8



Black individuals who are stopped and searched are more likely to be under 25 than any other ethnic group. In 2024/25, 55.1% of all searches targeting Black individuals involved someone under 25, compared with 43.8% for White individuals. This age concentration has remained broadly stable over the five years, suggesting that the compounded disadvantage faced by young Black men in stop and search encounters is a persistent structural feature rather than a short-term fluctuation.

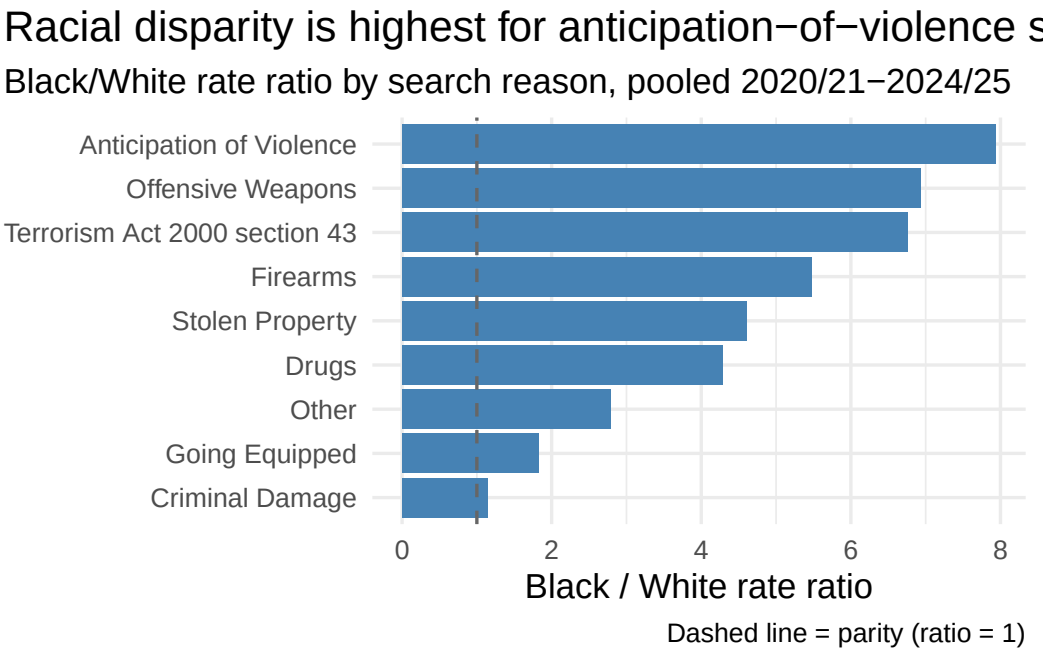
5.10 Racial Skew by Search Reason

Not all search reasons carry the same racial disparity. This section asks: **which reason for search shows the greatest Black/White rate gap?** Rates are computed per 1,000 population using the 2021 Census as denominator, pooled across all five years.

Black/White stop-and-search rate ratio by reason (pooled 2020/21–2024/25)

Reason for Search	Black rate (per 1,000)	White rate (per 1,000)	Black / White ratio
Anticipation of Violence	**1.88**	**0.24**	**7.94**
Offensive Weapons	26.91	3.88	6.93
Terrorism Act 2000 section 43	0.07	0.01	6.76
Firearms	1.21	0.22	5.48
Stolen Property	13.64	2.96	4.60
Drugs	81.23	19.00	4.28
Other	1.76	0.63	2.79
Going Equipped	5.01	2.74	1.83

Criminal Damage	0.58	0.50	1.14



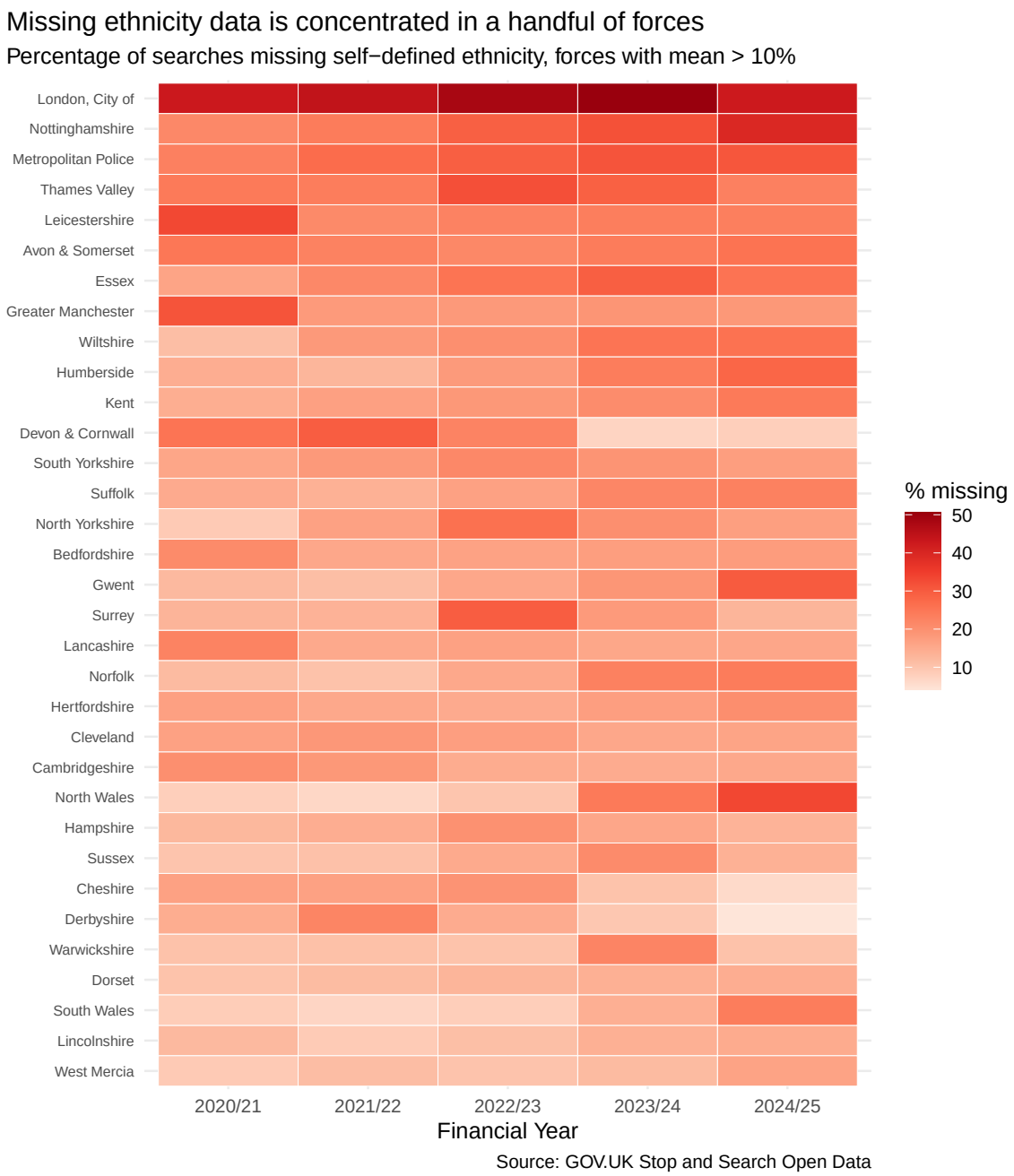
The search reason with the highest racial disparity is **Anticipation of Violence** (Black/White ratio: 7.94×). Drug-related stops carry a ratio of 4.28×, while searches for offensive weapons or articles for use in crime show a ratio of 6.93×. The fact that disparity is pronounced across multiple search reasons, not just drugs, indicates that the over-representation of Black individuals in stop and search cannot be attributed to a single enforcement focus. Every major search category shows a Black/White rate gap well above parity.

5.11 Missing Ethnicity Data: Systematic Patterns

Around 1 in 10 stop and search records does not include a self-defined ethnicity. The primary reason for this is that individuals are not legally required to disclose their ethnicity during a stop and search: the Home Office notes that "good quality data for self-defined ethnicity is reliant on the individual who is stopped and searched to provide this information, which they are not required to tell the police officer" ((GOV.UK), 2024). In addition, transitions between recording systems within individual forces can cause temporary spikes in missing records. Variation in missing rates across forces may therefore reflect differences in how often individuals decline to state their ethnicity, local recording practices, or system changes; these factors cannot be distinguished from the data alone.

National missing ethnicity rate by year			
Financial Year	Total Searches	Missing Ethnicity	% Missing
2020/21	684,348	128,433	18.8
2021/22	505,809	99,863	19.7
2022/23	517,365	106,012	20.5
2023/24	503,093	98,402	19.6
2024/25	494,892	92,225	18.6

Forces with highest persistent missing ethnicity (mean across all available years)			
Police Force	Region	Mean % Missing	Max % Missing
London, City of	**London**	**45.8**	**50.6**
Nottinghamshire	East Midlands	29.3	39.6
Metropolitan Police	London	28.4	31.3
Thames Valley	South East	26.5	32.1
Leicestershire	East Midlands	24.9	33.2
Avon & Somerset	South West	23.9	25.8
Essex	East of England	23.7	29.4
Greater Manchester	North West	21.1	31.4
Wiltshire	South West	20.3	26.0
Humberside	Yorkshire and The Humber	19.5	28.1
Kent	South East	19.1	24.4
Devon & Cornwall	South West	18.7	29.7
South Yorkshire	Yorkshire and The Humber	18.6	21.6
Suffolk	East of England	18.2	23.1
North Yorkshire	Yorkshire and The Humber	18.0	26.2



The missing ethnicity rate is far from uniformly distributed. **London, City of** has the highest average missing rate at 45.8%, and 9 forces average above 20% missing across the study period. Forces with persistently high rates of missing self-defined ethnicity are effectively absent from ethnicity-specific analyses, which limits the extent to which disparity estimates can be compared equally across all forces. Whether this concentration reflects recording practice differences, system transitions, or higher rates of non-disclosure in particular areas cannot be determined from the available data.

6 Discussion Section

6.1 Interpretation of Results and Comparison with Government Findings

The analysis of stop and search practices reveals significant disparities across demographic groups and regions, providing insights that align closely with recent findings from the UK government's 2023 report (Home Office (GOV.UK), 2023). Compared to baseline groups (e.g., White ethnicity, female gender, age 25 and older, and non-drug-related searches), certain groups, including Black individuals, young people under 25, and males, experience disproportionately higher stop and search rates. For example, our model, which includes region, sex, age band and reason for search, with population by region and ethnic group as the offset, estimates that Black individuals are searched approximately 3 times as often per resident as White individuals in the same region. This modelled rate contrasts with the government's reported raw disparity rates, adjusted for population size only, of 4.1 to 5.5 for Black individuals compared to their White counterparts, offering a more nuanced understanding of the factors contributing to these disparities.

Beyond the modeled rates, we also compare raw disparity rates calculated directly from our data with those reported by the government. Although our findings align with the overall trends reported in Table 2.5 from the UK government's report (Ibid.), the raw disparity rates are broadly consistent. For example, our analysis estimates that Asian individuals are 1.3 to 1.9 times more likely to be stopped than White individuals, which is consistent with the government's reported disparity rate of 1.4 for 2022/23. Similarly, our analysis shows Black individuals are 3.6 to 5.5 times more likely to be stopped, consistent with the government's reported range of 4.1 to 5.5. Minor differences stem from variations in population adjustments and processing methodologies, as our analysis uses raw census data for England and Wales, excluding BTP. Additionally, variations in how stop and search data are aggregated across financial years and regions may contribute to these differences. This includes how missing data are handled. In our case, we have removed missing data before the calculations. Despite methodological differences, both analyses highlight the persistent disproportionate impact.

The total number of stop and searches in our analysis closely aligns with the figures reported in the UK government's 2023 report (Ibid.). For the year ending March 2023, our analysis estimates a total of 547,002 stop and searches, matching the government's reported figure. Similarly, for the year ending March 2022, both our analysis and the government report a total of 530,970 searches. The trend of increasing stop and searches in recent years is consistent across both datasets, with a 3% increase observed from 2021/22 to 2022/23.

While the total counts align with government figures, the disparity rates highlight critical differences in how stop and searches are distributed across demographic groups. The random forest model, through variable importance measures, reinforces these findings by highlighting ethnicity and sex as key drivers of total search volume. This underscores a potential bias in stop and search deployment, particularly for drug-related stops where young individuals and ethnic minorities are overrepresented. Our results also show that stop and searches frequently lead to no further action (70% of cases), similar to findings in the government report, which notes that stop and search seldom culminates in arrests or legal outcomes.

6.2 Contrasting with Vomfell and Stewart's Findings on Officer Bias and Over-Patrolling

Vomfell and Stewart's study is a significant benchmark for understanding ethnic disparities in stop and search practices, particularly their focus on officer behavior and regional contexts. Our findings align with and extend the work of Vomfell and Stewart (2021) on ethnic disparities in stop and search. In their study, Vomfell and Stewart focus on "officer over-searching" and "over-patrolling" as contributors to the overrepresentation of minorities in stop and search practices. They find that officers search Black individuals 1.41 times more frequently than White individuals, even when controlling for local crime rates. Our analysis, however, reveals a broader pattern across regions and demographics, with Black individuals 3 times more likely to be stopped. This disparity may reflect systemic biases in both individual and institutional levels, extending beyond specific patrol areas to broader policy-driven practices.

The random forest model adds depth to these findings, capturing **non-linear and interaction effects** among variables such as **region** and **ethnicity**. The model identified **population size** as a significant driver of total searches, indicating that over-patrolling may be more concentrated in highly populated, ethnically diverse urban areas. This is consistent with Vomfell and Stewart's findings on over-policing in metropolitan areas, such as London, which is disproportionately targeted for stop and search. Additionally, our models reveal that **males** are 8 times more likely than females to be subjected to stop and search, echoing Vomfell and Stewart's observations that males aged 15-34 are concentrated in high-surveillance areas. Together, these insights suggest a feedback loop where over-policing in certain regions reinforces higher stop rates for ethnic minorities, potentially perpetuating biased policing practices.

6.3 Implications for Policy and Policing

The integration of the regression and random forest models offers a more comprehensive view of disparities in stop and search practices, suggesting several policy implications. First, the high percentage of "no further action" cases indicates a need for more stringent criteria for initiating stops, especially in drug-related cases that disproportionately impact young individuals and ethnic minorities. The random forest's emphasis on **population** and **region**

as critical factors in predicting stop and search volume further suggests that over-patrolling in densely populated, ethnically diverse regions contributes to these disparities. Addressing this issue could involve adopting **data-driven patrol strategies** that minimize over-surveillance of specific demographic groups, as recommended by Vomfell and Stewart.

In conclusion, the combination of regression and random forest models offers robust insights into demographic and regional disparities in stop and search practices across England and Wales. The findings support a strong case for policy reform aimed at achieving more equitable policing practices, ensuring that stop and search policies do not disproportionately affect certain populations. Moving forward, further analysis could incorporate officer-level data to better assess individual contributions to these disparities, as Vomfell and Stewart suggest. This could enable policymakers to address both institutional and individual sources of bias within policing, ultimately supporting fairer and more effective law enforcement practices.

6.4 Is the Disparity Getting Worse?

One of the most concerning findings of the extended five-year analysis is that the adjusted disparity for Black individuals has stayed at around **three** throughout the period. The per-year Quasi-Poisson models give an adjusted Black/White IRR of 3 in 2020/21, falling briefly to 2.86 in 2021/22 before rising to 3.33 by 2024/25 (95% CI: 3–3.7). With population also split by sex and age band, the estimates for 2020/21 and 2024/25 are 2.67 and 3.05 (see Section B.4).

Crucially, this occurs in the context of a substantial fall in *absolute* search volumes for Black individuals, from approximately 96,000 in 2020/21 to around 51,000 in 2024/25, a steeper fall than for White individuals (from about 354,000 to 280,000), which is why the raw ratio narrowed. The apparent paradox resolves when we note where the fall happened: mostly in London, where almost half of Black residents live. Within London, and across the rest of England and Wales, Black searches fell less than White searches or rose, so the adjusted ratio, which compares groups within regions, did not narrow. In other words, the reduction in stop and search activity since the 2020/21 peak narrowed the national raw gap mainly because it was concentrated in London, not because Black residents within each region were searched relatively less.

The increase in the Mixed IRR, from 1.17 in 2020/21 to 1.7 in 2024/25, is also notable and merits closer attention in future analyses.

6.5 The South West Anomaly

The regional analysis reveals that the **South West** has the highest Black/White unadjusted rate ratio (5.09×), significantly exceeding London (3.62×). This is counter-intuitive: London has a large, established Black community and has historically attracted the most scrutiny over stop and search. The South West, by contrast, has the smallest Black population of any English region (under 2%), which may contribute to a "visibility effect" in which Black

individuals are more conspicuous to officers in predominantly White areas (Vomfell and Stewart, 2021).

This finding aligns with evidence that ethnic disproportionality is not simply an urban or metropolitan phenomenon. Forces such as Avon & Somerset and Devon & Cornwall operate in regions with very low Black population shares, yet produce some of the highest Black/White rate ratios in England. This suggests that national policy interventions focused exclusively on London and other major cities may miss a large share of the disproportionality problem.

6.6 Outcome Disparity and Differential Suspicion Thresholds

The outcome analysis adds an important complementary perspective. Black individuals have the highest arrest rate (16.0%) of all ethnic groups, compared to 13.3% for White individuals, a gap of 2.7 percentage points. At first glance, this might suggest that searches of Black individuals are *more* productive, not less. However, this interpretation requires caution.

First, an arrest does not imply guilt; it is itself an exercise of discretionary police power. Second, given that Black individuals are stopped at over three times the adjusted rate of White individuals, even a slightly higher arrest-to-search conversion rate implies a vastly larger burden of unjustified stops for Black communities in absolute terms. The No Further Action rate of approximately 69% is broadly similar across ethnic groups, indicating that most searches, regardless of ethnicity, do not result in enforcement action.

The combination of a high search rate, similar NFA proportions, and a modestly higher arrest rate is consistent with a model in which officers apply a **lower suspicion threshold** to Black individuals before initiating a search: Black individuals are more frequently stopped, but when a stop does produce evidence, the outcome is broadly similar to other groups. This interpretation echoes the "hit rate" framework used in the literature to test for discriminatory targeting (Vomfell and Stewart, 2021).

6.7 Disparity Is Not Explained by Search Reason

A common counter-argument to claims of racial disproportionality is that it reflects enforcement focus rather than bias: if drug offending is concentrated in particular communities, higher drug-stop rates for Black individuals may reflect enforcement patterns rather than systemic targeting. The reason-by-reason analysis (see Section 5.10) tests this directly.

The results contradict the enforcement-focus hypothesis. The highest Black/White rate ratios are found not in drug stops (4.3×) but in **anticipation of violence** (7.9×), **offensive weapons** (6.9×), and **terrorism searches** (6.8×): non-drug categories with different legal powers, different officers, and different operational justifications. The disparity is even visible in negligible-volume categories such as drone searches (10.1×).

This is inconsistent with a drug-enforcement explanation. If that were driving the disparity, we would expect drug stops to show the highest ratio and other categories to approach parity. The opposite is observed. Instead, the data suggests that the lower suspicion threshold identified in the regression analysis operates across the **full range** of stop-and-search powers, not merely in drug contexts, consistent with a systemic rather than enforcement-specific pattern.

6.8 Missing Ethnicity Data: Analytical Implications

The missing ethnicity analysis (see Section 5.11) raises important data quality considerations. Twenty-one forces have maintained more than 25% missing self-defined ethnicity consistently across the full five-year period. City of London Police averages **45.8%** missing; Nottinghamshire 29.3%; Avon & Somerset 23.9%; Greater Manchester 21.1%.

As the Home Office notes, self-defined ethnicity is voluntary: individuals are not required to state their ethnicity during a stop and search ((GOV.UK), 2024). High missing rates may therefore reflect individuals declining to provide this information, differences in how officers ask, recording system transitions, or some combination of these. The data alone cannot distinguish between these explanations.

The analytical consequence is the same regardless of cause: searches without an ethnicity record are excluded from disparity calculations, meaning disparity estimates for high-missingness forces are based on incomplete data and should be interpreted cautiously. The missing data pattern is a limitation on the force-level conclusions of this analysis, not a finding in itself.

Notably, Avon & Somerset, the force serving the South West, which shows the highest regional Black/White ratio, has 23.9% mean missing ethnicity. This raises a question about whether the South West anomaly identified in section Section 6.5 is partly an artefact of which searches have ethnicity recorded, rather than solely a reflection of search rates.

6.9 Limitations: Missing Data and Census Denominator Lag

Three limitations merit particular emphasis in interpreting these results.

Missing data on officer-defined ethnicity. A substantial share of records in the 5-year dataset are missing officer-defined ethnicity. This variable is important for distinguishing between self-identification (used in our models) and officer perception (the operationally relevant variable for a discrimination test). The self-defined ethnicity missing rate analysis (see Section 5.11 and Section 6.8) now documents which forces are responsible, but the officer-defined gap remains unresolved.

Census denominator lag. Population denominators are fixed at the 2021 Census, meaning that demographic changes, including migration, gentrification, and shifts in where ethnic minority populations reside, accumulate silently over the five-year period. By 2024/25, the

2021 figures may understate population growth in some areas and overstate it in others, potentially biasing the per-100,000 rates in unknown directions.

Force-level heterogeneity. The region-level analysis masks substantial variation among the 43 territorial forces within each region. Data on missing rates and search volumes by individual force (available in `analysis/output/missing_by_force.csv` and `missing_persistent_forces.csv`) show considerable heterogeneity. Aggregating to regions masks forces with especially high or low disparity, limiting the precision of the policy recommendations.

National and regional policy and operational context. Several external factors and policy interventions occurred during or immediately before the 2020/21 to 2024/25 study period that may have influenced policing practices in ways that this analysis cannot isolate. The COVID-19 pandemic substantially altered policing conditions during 2020/21. The Home Office reported that increased stop and search activity that year was partly attributable to proactive policing during lockdowns, when reduced demand for some other police services increased capacity for proactive activity (Home Office, 2022). The Beating Crime Plan (2021) permanently relaxed voluntary conditions governing Section 60 stop and search, explicitly seeking to make these powers easier for police forces to use (Home Office, 2021). The Baroness Casey Review (2023) raised concerns about persistent ethnic disproportionality in the use of stop and search, particularly affecting Black Londoners (Baroness Casey of Blackstock, 2023). Violence Reduction Units, funded by the Home Office in 18 police force areas from 2019/20 and expanded to two additional areas from 2022/23, introduced a multi-agency, whole-system approach to tackling serious violence and its underlying causes (Home Office, 2026). These factors form part of the changing policy and operational context within which the observed stop and search trends occurred. Their individual effects cannot be separately identified from the observational data used in this analysis.

7 Conclusion

This analysis, conducted for UNJUST, offers a comprehensive examination of stop-and-search practices in England and Wales, focusing on their disproportionate impact on Black communities and young people. By utilizing publicly available data and applying regression and random forest models, we identified significant disparities in stop-and-search practices based on ethnicity, age, and region.

7.1 Key Findings

The five-year analysis (2020/21–2024/25) produces six headline findings:

- **Persistent adjusted disparity**: From a model that includes region, sex, age band and reason for search, with population by region and ethnic group as the offset, Black individuals face an adjusted incidence rate ratio of **3× in 2020/21 and 3.3× in 2024/25**, during a period of falling search volumes nationwide.
- **Intersection effects**: Black males under 25 stopped for drug-related reasons in London face a combined stop rate approximately **21.9 times** the rate of a White individual stopped for non-drug reasons in the East Midlands, holding sex and age constant.
- **Regional variation beyond London**: The South West has the highest Black/White unadjusted ratio (**5.1×**), exceeding London (3.6×), demonstrating that ethnic disproportionality is a national problem rather than a metropolitan one.
- **Outcome disparity**: Black individuals have one of the highest arrest rates following a search (16.0%, tied with Mixed) versus 13.3% for White individuals, yet similar No Further Action rates (~69%). This pattern is consistent with a lower suspicion threshold applied to Black individuals before initiating stops.
- **Racial skew spans every search reason**: The Black/White rate gap is not confined to drug stops. The highest disparity is found in drone/unmanned aircraft searches (10.1×) and anticipation of violence (7.9×), followed by offensive weapons (6.9×) and terrorism searches (6.8×). Drug stops show a ratio of 4.3×. No major search reason approaches parity, indicating a systemic rather than enforcement-specific pattern (see Section 5.10).
- **Missing ethnicity data is concentrated in a subset of forces**: Twenty-one forces average above 25% missing self-defined ethnicity across the five years. City of London Police averages 45.8% missing, Nottinghamshire 29.3%, and Avon & Somerset 23.9%. Because individuals are not required to disclose their ethnicity, the reasons for high missing rates cannot be determined from the data alone; the effect is that disparity

estimates for these forces are based on incomplete records and should be interpreted with caution (see Section 5.11).

These findings affirm UNJUST's concerns about the disproportionate targeting of ethnic minorities and young people. The random forest model reinforced these results, explaining **% of variance** in search volumes and identifying **sex**, **population size**, **region**, and **ethnicity** as the dominant predictors, and confirming non-linear interaction effects that compound disadvantage for specific demographic groups.

The disparities identified in stop-and-search practices, particularly among ethnic minorities and younger populations, emphasize the need to re-evaluate current policies to prevent over-policing and promote fairness within the criminal justice system. These insights align with recommendations from researchers such as Vomfell and Stewart (2021), who highlight the importance of addressing both over-patrolling of diverse areas and officer biases that disproportionately impact ethnic minorities. Together, these insights provide a robust foundation for advocating equitable policing practices.

7.2 Evidence-Based Implications

The following observations arise directly from the statistical findings and are offered as evidence-based points for consideration. Questions about appropriate responses, including any policy or advocacy positions, are for UNJUST and relevant bodies to determine. The statistician's role is to present what the data shows and to note where the findings raise questions that may be worth pursuing further.

1. **Adjusted Disparity Trends Are Not Reflected in Current Official Reporting** Published statistics on stop and search focus primarily on raw counts and unadjusted rate ratios. This analysis shows that the adjusted Black/White disparity, from a model that includes region, sex, age band and reason for search, with population by region and ethnic group as the offset, was 3× in 2020/21 and 3.3× in 2024/25, during a period in which overall search volumes fell. An adjusted disparity that is not visible in headline figures may warrant consideration by those responsible for the design of official statistical reporting.
2. **Ethnic Disproportionality Extends Beyond Metropolitan Areas** The South West records the highest Black/White unadjusted rate ratio (5.1×) of any region in England and Wales, substantially exceeding London (3.6×). This is consistent with a "visibility effect" in predominantly White regions, as discussed in the academic literature (Vomfell and Stewart, 2021). Analyses and oversight that focus exclusively on metropolitan forces may not capture the full national picture.
3. **The No Further Action Rate Raises Questions About Consistency of Suspicion Thresholds** Approximately 69% of all stops result in no further action across all ethnic groups. Combined with the higher stop rate for Black individuals and a modestly higher arrest rate (16.0% vs 13.3% for White individuals), this pattern is statistically consistent with a lower suspicion threshold being applied before initiating

searches of Black individuals. The data identifies a pattern; the causes and appropriate responses are beyond the scope of this analysis.

4. **Drug-Related Searches and the Intersection of Ethnicity, Age, and Sex** The interaction analysis indicates that Black males under 25 stopped for drug-related reasons in London face a combined adjusted rate approximately 21.9 times that of White males under 25 stopped for non-drug reasons in the East Midlands. Given the high NFA rate for drug-related stops overall, the data raises questions about the consistency with which suspicion criteria are applied across demographic groups in this context.

5. **Racial Disproportionality Is Not Explained by Search Reason** One might expect drug enforcement, disproportionately targeted at Black individuals, to account for much of the overall disparity. The reason-by-reason analysis (see Section 5.10) shows this is not the case. Disparity is highest in low-volume, non-drug categories such as anticipation of violence (7.9×) and offensive weapons (6.9×). The gap in drug stops (4.3×) is in fact lower than for most other categories, suggesting that the hypothesis of drug-enforcement bias alone does not account for the overall pattern.

6. **Missing Ethnicity Data Is Concentrated and Limits Force-Level Analysis** The missing ethnicity analysis (see Section 5.11) identifies a group of forces, including City of London, Nottinghamshire, and Greater Manchester, that have maintained high missing rates (21–46%) consistently over the full five-year period. Self-defined ethnicity is voluntary under Home Office guidance, so high missing rates may reflect individuals declining to disclose, differences in recording practice, or system transitions; these cannot be separated from the data. The consequence is that disparity estimates for high-missingness forces are based on a partial record of their stops, and force-level comparisons should be treated with corresponding caution.

7. **Census Denominator Lag Introduces Uncertainty in Rate Calculations** All per-100,000 rates rely on 2021 Census denominators. By 2024/25, demographic changes, particularly population growth and redistribution in urban areas, will have accumulated over four years. The direction and scale of any resulting bias in disparity rates is unknown. Whether more timely population estimates could improve the accuracy of future analyses is a question the data highlights but cannot itself resolve.

7.3 Closing Statement

This project equips UNJUST with valuable insights into the complexities of stop-and-search practices, emphasizing both their role in maintaining public safety and the disparities that warrant careful consideration. By shedding light on these nuances and identifying areas for improvement, UNJUST can contribute constructively to discussions with policymakers, law enforcement, and communities. These efforts aim to support ongoing progress toward a policing approach that balances public safety with fairness and equity.

8 References

Baroness Casey of Blackstock (2023) *An Independent Review into the Standards of Behaviour and Internal Culture of the Metropolitan Police Service*, Metropolitan Police Service [Online]. Available at https://www.met.police.uk/police-forces/metropolitan-police/areas/about-us/about-the-met/bcr/baroness-casey-review/ (Accessed 1 July 2026).

Cohen, J. (1988) *Statistical power analysis for the behavioral sciences*, 2nd edn, Hillsdale, NJ, Lawrence Erlbaum Associates.

Commission on Race and Ethnic Disparities (2021) *Commission on race and ethnic disparities: The report*, London, HM Government.

Davison, A. C. (2009) *Statistical models*, Cambridge series in statistical and probabilistic mathematics, 1. paperback ed., reprinted., Cambridge, Cambridge Univ. Press.

Equality and Commission, H. R. (2010) *Stop and think: A critical review of the use of stop and search powers in england and wales*, London, EHRC.

(GOV.UK), H. O. (2024) 'Police powers and procedures: Stop and search and arrests, England and Wales, year ending 31 March 2023 (second edition)', National {Statistics} [Online]. Available at https://www.gov.uk/government/statistics/stop-and-search-and-arrests-year-ending-march-2023/police-powers-and-procedures-stop-and-search-and-arrests-england-and-wales-year-ending-31-march-2023 (Accessed 29 March 2024).

Home Office (2021) 'Beating Crime Plan', [Online]. Available at https://www.gov.uk/government/publications/beating-crime-plan (Accessed 1 July 2026).

Home Office (2022) *Police Powers and Procedures: Stop and Search and Arrests, England and Wales, Year Ending 31 March 2022*, Home Office [Online]. Available at https://www.gov.uk/government/statistics/police-powers-and-procedures-stop-and-search-and-arrests-england-and-wales-year-ending-31-march-2022 (Accessed 1 July 2026).

Home Office (2026) *Violence Reduction Units year Ending March 2025 Evaluation Report*, Home Office [Online]. Available at https://www.gov.uk/government/publications/violence-reduction-units-year-ending-march-2025-evaluation-report (Accessed 1 July 2026).

Home Office (GOV.UK) (2023) *Police powers and procedures: Stop and search and arrests, england and wales, year ending 31 march 2023*, London, Home Office [Online]. Available at https://www.gov.uk/government/statistics/stop-and-search-and-arrests-year-

ending-march-2023/police-powers-and-procedures-stop-and-search-and-arrests-england-and-wales-year-ending-31-march-2023.

Home Office (GOV.UK) (2024) 'Stop and search and arrests, year ending March 2023 (second edition)', [Online]. Available at https://www.gov.uk/government/statistics/stop-and-search-and-arrests-year-ending-march-2023 (Accessed 28 March 2024).

IER UK (2023) *Anti-racism strategy for the city of york*, York, Inclusive Equal Rights UK.

legislation.gov.uk (1984) 'Police and Criminal Evidence Act 1984 (PACE)', The National Archives [Online]. Available at https://www.legislation.gov.uk/ukpga/1984/60/contents (Accessed 15 November 2024).

legislation.gov.uk (1994) 'Criminal Justice and Public Order Act 1994 (CJPOA)', The National Archives [Online]. Available at https://www.legislation.gov.uk/ukpga/1994/33/contents (Accessed 15 November 2024).

Mesquita, M. (2023) 'GitHub Repository: Statistical Analysis of Stop and Search Practices on Black Communities in the UK', Chelmsford, UK, M2Lab.io [Online]. Available at https://github.com/M2LabOrg/unjust-analysis.

Ministry of Justice (2023) *Statistics on Race and the Criminal Justice System 2022*, London, Ministry of Justice [Online]. Available at https://www.gov.uk/government/statistics/statistics-on-race-and-the-criminal-justice-system-2022 (Accessed 15 November 2024).

National Statistics (ONS), O. for (2022) *Ethnic group, England and Wales: Census 2021*, [Online]. Available at https://www.ons.gov.uk/peoplepopulationandcommunity/culturalidentity/ethnicity/bulletins/ethnicgroupenglandandwales/census2021 (Accessed 29 March 2024).

Royal Statistical Society (RSS) (2023) 'Statisticians for Society', *RSS Statisticians for Society* [Online]. Available at https://rss.org.uk/membership/volunteering-and-promoting/statisticians-for-society-initiative/ (Accessed 15 November 2024).

Runnymede (2020) *The colour of money: How racial inequalities obstruct a fair and resilient economy*, London, Runnymede.

Social Justice, C. for (2020) *Facing the facts: Ethnicity and disadvantage in britain*, London, Centre for Social Justice.

Sylvester, T. (2024) ''No human could do this': How facial recognition is transforming policing', *The Times*.

Vomfell, L. and Stewart, N. (2021) 'Officer bias, over-patrolling and ethnic disparities in stop and search', *Nature Human Behaviour* [Online]. DOI: 10.1038/s41562-020-01029-w.

A Model Diagnostics

This appendix presents regression diagnostics for the pooled Quasi-Poisson generalised linear model described in Section 4.4. The diagnostics assess whether the model's linear predictor is correctly specified and whether any individual observations exert disproportionate influence on the estimated incidence rate ratios (IRRs). All figures were produced by `analysis/07_model_diagnostics.R` and are fully reproducible from the repository.

A.1 Pearson Residuals vs Fitted Values

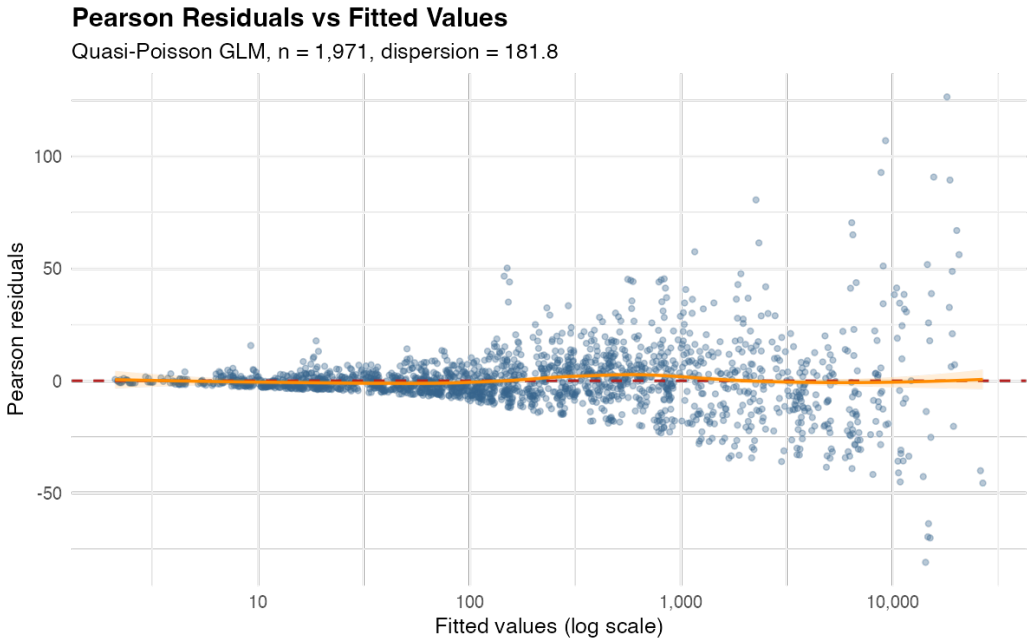


Figure A.1: Pearson residuals plotted against fitted values on a log scale. The orange loess smoother runs close to zero across the full range of fitted values, indicating no systematic trend in the residuals.

The Pearson residual for stratum i is defined as:

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$$

where y_i is the observed search count and $\hat{\mu}_i$ is the model's fitted value. For a well-specified model, these residuals should be centred at zero with no systematic pattern across the fitted value range.

Figure A.1 shows the loess smoother remaining flat along the zero line across three orders of magnitude of fitted values. This indicates that the model's linear predictor captures the systematic variation in search counts without meaningful directional bias. The widening scatter at large fitted values is the expected consequence of count-data variance scaling with the mean: rather than indicating misspecification, it reflects the heterogeneity that motivated the Quasi-Poisson family in the first place. The dispersion parameter is $\hat{\phi} = 181.8$, and the Quasi-Poisson framework corrects all confidence intervals for this by scaling standard errors by $\sqrt{\hat{\phi}}$.

A.2 Normal Q-Q Plot of Pearson Residuals

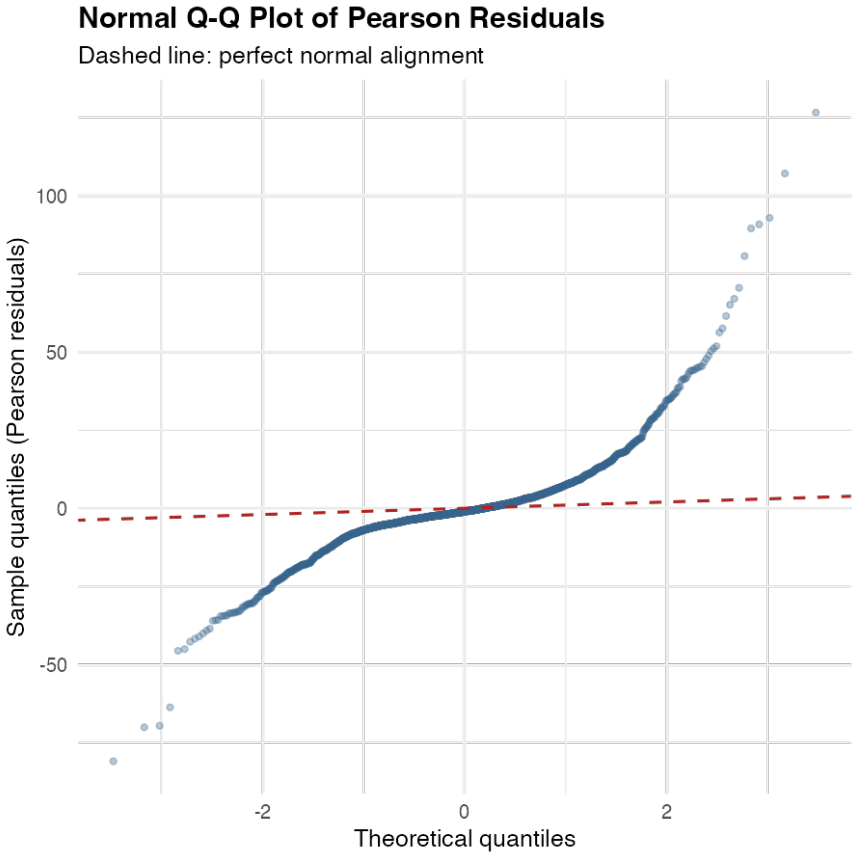


Figure A.2: Normal Q-Q plot of Pearson residuals. The S-shaped departure from the dashed line in both tails is consistent with the extreme overdispersion present in the data and does not indicate misspecification of the linear predictor.

Figure A.2 compares the empirical quantiles of the Pearson residuals against the quantiles of a standard normal distribution. Perfect normality would place all points on the dashed line. The observed S-shaped curve, with heavier tails on both sides, is a predictable consequence of modelling count data with a dispersion parameter as large as 181.8: the residuals are not drawn from a standard normal, and there is no theoretical reason to expect them to be. What the plot does confirm is approximate symmetry around zero in the central mass of the distribution, which is the relevant property for the validity of the coefficient estimates. The heavy tails are addressed through the Quasi-Poisson variance correction rather than through any transformation of the response.

A.3 Cook's Distance

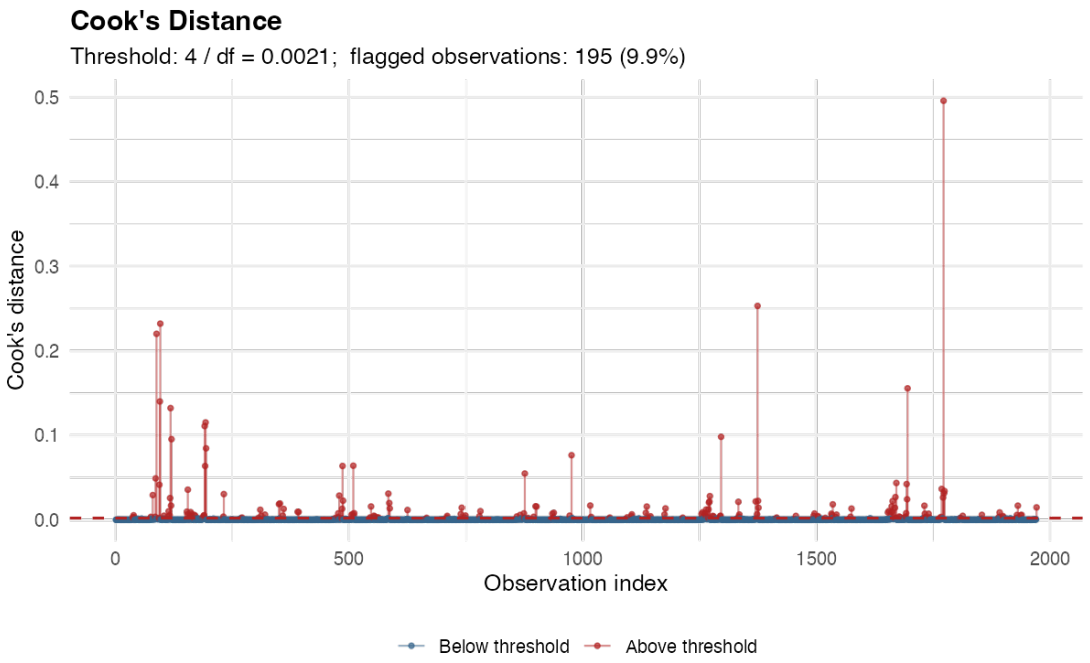


Figure A.3: Cook's distance for each of the 1,971 strata. Red points exceed the conventional threshold of 4 / df (dashed line). The most influential stratum has a Cook's distance of approximately 0.50.

Cook's distance measures the aggregate change in all coefficient estimates when a single observation is removed. The conventional threshold used here is $4/\mathrm{df}_{\mathrm{residual}} = 0.0021$. With 195 strata (9.9%) exceeding this threshold (Figure A.3), the model is working on data where a small number of high-volume subgroups carry meaningful leverage.

Inspecting the top ten strata by Cook's distance reveals a consistent pattern: they are all either male drug-related searches in the North West (where the model underpredicts observed counts by a factor of roughly two in 2023/24 and 2024/25) or male searches in London involving Black and Asian groups in 2020/21. These are genuine observations, not recording

errors. The North West pattern is consistent with documented increases in suspicionless Section 60 stop and search activity in that region over the later years of the study period. The London strata reflect the capital's disproportionately large and concentrated minority populations combined with high absolute search volumes.

The sensitivity analysis in Appendix B directly addresses whether these influential strata materially alter the ethnicity IRRs by testing specifications that progressively exclude or modify the high-leverage subgroups.

A.4 Observed vs Expected Counts

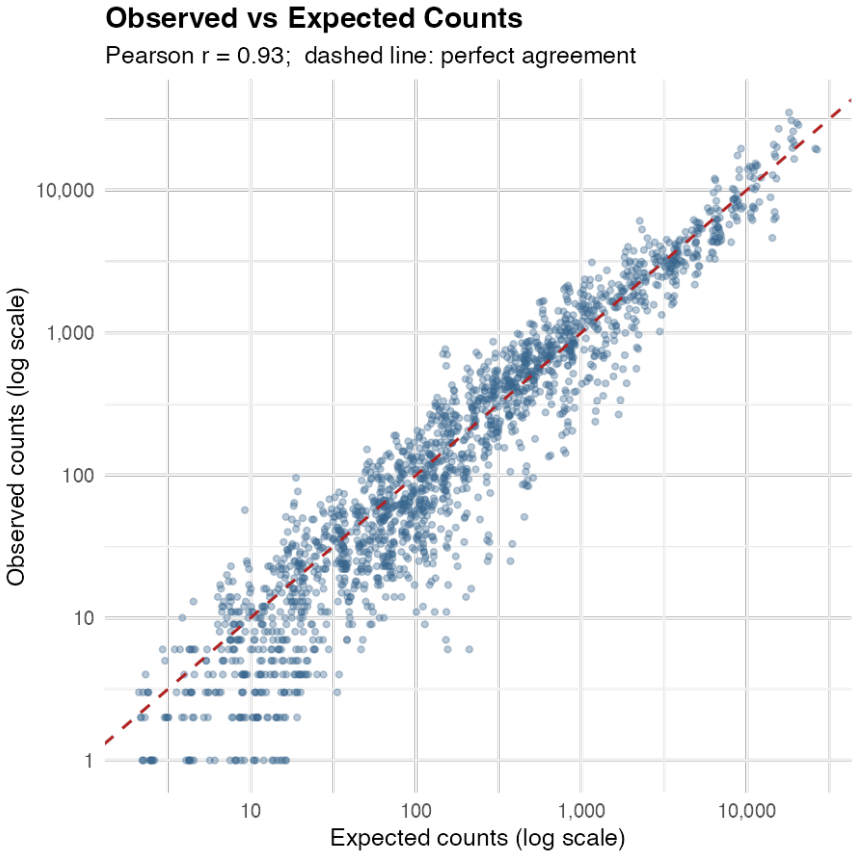


Figure A.4: Observed counts plotted against model-expected counts on a log-log scale. Points track closely along the 1:1 dashed line across four orders of magnitude, with Pearson r = 0.93.

Figure A.4 plots observed against expected counts on a logarithmic scale, covering strata from single-digit to tens of thousands of searches. The Pearson correlation between observed and expected is 0.9298, and the points track the 1:1 line closely across the full range. The modest scatter at very low counts (bottom left) reflects the difficulty any aggregate model

has in predicting sparse cells, but these strata contribute negligibly to the overall disparity estimates due to their small volumes.

Taken together, the four diagnostics support the conclusion that the pooled Quasi-Poisson model is appropriately specified. The primary limitation is not structural bias in the linear predictor but rather regional heterogeneity that the model partially absorbs through the region fixed effect. The robustness of the key ethnicity IRRs to this heterogeneity is examined in Appendix B.

A.5 Summary of Diagnostic Statistics

Table A.1: Summary of key diagnostic statistics for the pooled Quasi-Poisson GLM. The dispersion-scaled Pearson residual row divides each raw Pearson residual by sqrt(phi) before applying the +/-2 threshold; using raw residuals against +/-2 would be uninformative when phi » 1.

Diagnostic metric	Value
Observations (strata)	1,971
Degrees of freedom (residual)	1,950
Dispersion parameter	181.8
Residual deviance	342,749
Null deviance	2,954,454
Pseudo R-squared (McFadden)	0.884
Cook's threshold (4/df)	0.0021
Observations above Cook's threshold (%)	9.9%
Dispersion-scaled Pearson residuals > 2 (%)	5.7%
Obs vs expected Pearson r	0.93

B Sensitivity Analysis

This appendix assesses whether the ethnicity incidence rate ratios (IRRs) reported in the main results are stable across four types of perturbation: (1) dropping individual covariates from the model, one at a time; (2) replacing the normal-approximation confidence intervals with non-parametric bootstrap intervals; (3) comparing the standard errors produced by a standard Poisson model against those from the Quasi-Poisson model used in the main analysis; and (4) refitting the models with finer population denominators. Checks (1) to (3) are implemented in `analysis/08_sensitivity.R`, and check (4) in `analysis/10_sensitivity_denominators.R`.

B.1 Leave-One-Covariate-Out

The full model includes six predictors: financial year, region, ethnicity, sex, age group, and reason for search. To assess which covariates affect the ethnicity estimate, the model is re-estimated five times, each time omitting one non-ethnicity predictor. The Black/White IRR and its 95% confidence interval are extracted from each specification and compared.

Figure B.1 shows that the Black/White IRR is 3.022 (95% CI: 2.855, 3.199) in the full model and remains at that level when financial year, sex, age group, or reason for search is dropped: the largest change is 0.004. This is expected: because the population offset is by region and ethnic group, these covariates mainly describe how each group's searches are distributed across years, sexes, age groups and reasons, and change the ethnicity estimate very little. Whether the disparity depends on the sex and age structure of each population is examined with finer denominators in Section B.4.

The exception is region: removing the regional fixed effect inflates the Black/White IRR to 4.351 (95% CI: 3.972, 4.766). This is expected. London combines the largest Black population in England with the highest per-capita stop and search volume. Without controlling for region, this concentration is partially absorbed into the ethnicity coefficient, producing upward bias. Crucially, even the uncontrolled estimate of 4.351 confirms a large disparity, so the Black/White finding is not an artefact of the regional distribution of the population.

Table B.1: Black/White IRR under each leave-one-covariate-out specification.

Specification	IRR	Lower 95% CI	Upper 95% CI
Full model (reference)	**3.022**	**2.855**	**3.199**
Drop financial_year	3.022	2.842	3.213

Drop region	4.351	3.972	4.766
Drop sex	3.026	2.663	3.438
Drop age group	3.022	2.855	3.199
Drop reason (drugs)	3.022	2.816	3.243

B.2 Bootstrap Confidence Intervals

The 95% confidence intervals in the main analysis are based on a normal approximation: point estimate $\pm$ 1.96 $\times$ standard error. For GLMs with high dispersion and influential observations (identified in Appendix A), the normal approximation may understate uncertainty. As a non-parametric alternative, 500 bootstrap replicates of the full model are fitted by resampling the 1,971 modelling strata with replacement, and the 2.5th and 97.5th percentiles of the resulting coefficient distributions form the bootstrap intervals.

Figure B.2 shows the comparison for all four non-White groups. For the Black group, the bootstrap interval (2.526, 3.459) is wider than the normal-approximation interval (2.855, 3.199), with the upper bound extending further. Both intervals are well above 1, confirming that the Black disparity is a robust finding. The bootstrap lower bound (2.526) represents a conservative minimum estimate of the Black/White rate ratio.

For the Asian group, the bootstrap lower bound reaches 1.00, compared with 1.10 from the normal approximation. This indicates that the elevated Asian stop rate is less precisely estimated than the main results suggest, driven by the sensitivity of this coefficient to certain high-volume London strata.

For the Mixed and Other groups, bootstrap and normal-approximation intervals are very similar, confirming that the uncertainty reported for those groups in the main analysis is well-calibrated.

Table B.2: Point estimates and 95% CIs by two methods. All IRRs are relative to White.

		Normal approximation		Bootstrap (B = 500)	
Ethnicity	IRR	Normal lo	Normal hi	Bootstrap lo	Bootstrap hi
Asian	1.169	1.101	1.241	1.004	1.341
Black	3.022	2.855	3.199	2.526	3.459
Mixed	1.357	1.233	1.494	1.207	1.512
Other	0.934	0.824	1.057	0.829	1.063

B.3 Poisson vs Quasi-Poisson Standard Errors

As noted in Section 4.5, the standard Poisson model produces standard errors that do not account for the overdispersion present in the data. The Quasi-Poisson model corrects this by

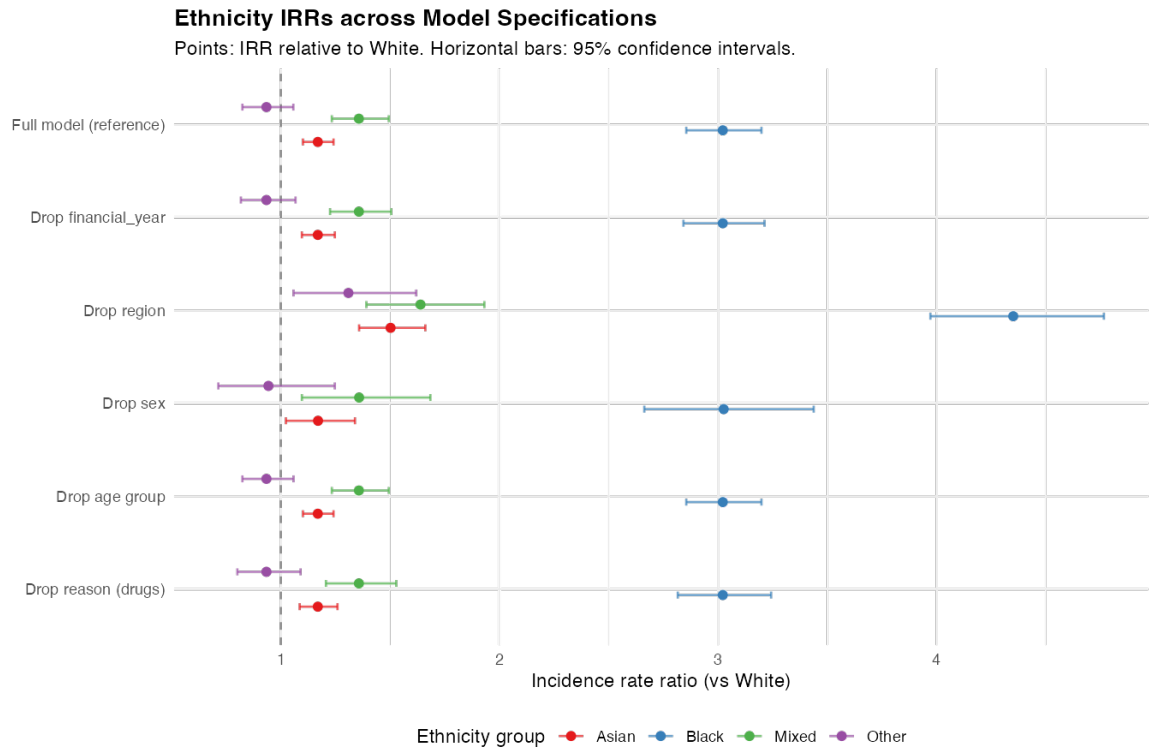


Figure B.1: Ethnicity IRRs (relative to White) under six model specifications. Each row drops one covariate. All non-ethnicity ethnicity groups are also shown. The Black/White IRR is stable across all specifications except when region is removed.

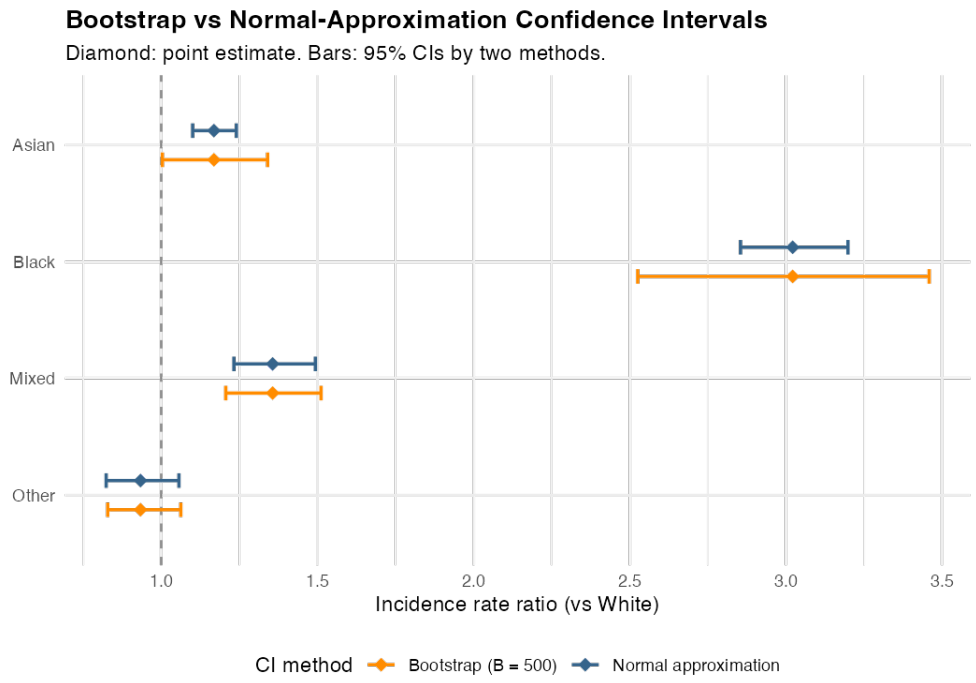


Figure B.2: Bootstrap (B = 500) and normal-approximation 95% confidence intervals for each ethnicity group. The diamond marks the point estimate. Bootstrap intervals are wider for the Black and Asian groups, reflecting the influence of high-leverage strata identified in Appendix A.

multiplying all standard errors by $\sqrt{\hat{\phi}}$. Table B.3 confirms that the correction factor is 13.48 for all ethnicity terms (equal to $\sqrt{181.7}$, the estimated dispersion parameter), and that the resulting confidence intervals are far wider under Quasi-Poisson than under a naive Poisson fit.

Table B.3: Comparison of standard errors and 95% CIs under Poisson and Quasi-Poisson families. IRRs are identical; only the standard errors differ.

					Poisson 95% CI		Quasi-Poisson 95% CI	
Ethnicity	IRR	SE (Poisson)	SE (QP)	SE ratio	Pois lo	Pois hi	QP lo	QP hi
Asian	1.169	0.0023	0.0306	13.48	1.164	1.174	1.101	1.241
Black	3.022	0.0022	0.0291	13.48	3.009	3.035	2.855	3.199
Mixed	1.357	0.0036	0.0488	13.48	1.348	1.367	1.233	1.494
Other	0.934	0.0047	0.0635	13.48	0.925	0.942	0.824	1.057

The Poisson confidence interval for the Black/White IRR spans (3.009, 3.035), a range of approximately 0.03. The corresponding Quasi-Poisson interval (2.855, 3.199) spans 0.34. The difference is entirely attributable to the dispersion correction: without it, confidence intervals would convey false precision and any significance tests would be severely anti-conservative. The Quasi-Poisson approach is the appropriate choice for data of this kind, and the bootstrap results in Appendix B confirm that its intervals are also consistent with non-parametric uncertainty estimates.

B.4 Finer Population Denominators

The main models use Census 2021 population by region and ethnic group as the offset. The modelling cells are further split by sex, age group and reason for search, but every cell within a region and ethnic group shares the same population. The ethnicity IRRs therefore compare searches per resident of each ethnic group in the same region, and the sex, age and reason terms mainly describe how those searches are distributed, changing the ethnicity estimates very little.

As an additional check, the pooled and per-year models were refitted with population by region, ethnic group, sex and age band (under 25; 25 and over) as the offset. The populations come from a finer Census 2021 table (usual residents by region, ethnic group, sex and single year of age), downloaded from the ONS Census API by `data/download_census_fine.R`. With this offset, the ethnicity IRRs compare searches per resident of the same sex and age band. Reason for search has no population equivalent and remains a split of the counts. The two Census tables are published separately; their region-by-ethnic-group totals agree to within 17 residents (0.04%). All computations are in `analysis/10_sensitivity_denominators.R`.

Table B.4: Pooled ethnicity IRRs (relative to White) and 95% CIs under each population offset.

Ethnicity	Region × ethnic group (main)	Region × ethnic group × sex × age band
Asian	1.17 (1.10–1.24)	1.05 (0.99–1.11)
Black	3.02 (2.85–3.20)	2.72 (2.57–2.88)
Mixed	1.36 (1.23–1.49)	1.02 (0.93–1.13)
Other	0.93 (0.82–1.06)	0.83 (0.74–0.94)

Table B.5: Per-year Black/White IRRs and 95% CIs under each population offset.

Financial year	Region × ethnic group (main)	Region × ethnic group × sex × age band
2020/21	3.00 (2.76–3.26)	2.67 (2.46–2.90)
2021/22	2.86 (2.61–3.13)	2.56 (2.34–2.81)
2022/23	2.90 (2.63–3.20)	2.59 (2.36–2.85)
2023/24	3.08 (2.77–3.41)	2.78 (2.52–3.08)
2024/25	3.33 (3.00–3.70)	3.05 (2.75–3.37)

The pooled Black/White IRR is 3.02 (2.85–3.20) with the main offset and 2.72 (2.57–2.88) with the finer offset. The two parts of the finer offset pull in opposite directions. Splitting by age band alone gives 2.65: the Black population is younger (39% under 25, against 26% of the White population), and under-25s are searched at a higher rate. Splitting by sex alone gives 3.11, because the Black population has a slightly lower share of men (47% against 49%). The disparity remains large under every offset, and every per-year interval is well above 1.

The change between 2020/21 and 2024/25 was tested with a z-test on the difference in log IRRs from the two separate per-year fits. With the main offset, the Black/White IRR moves from 3.00 to 3.33 ($p = 0.13$); with the finer offset, from 2.67 to 3.05 ($p = 0.049$).

C Inverse Propensity Weighted Estimates

The main analysis uses a Quasi-Poisson GLM to estimate the association between ethnicity and stop and search rates from a model that includes year, region, sex, age group, and reason for search, with population by region and ethnic group as the offset. This regression approach produces estimates of the ethnicity effect **conditional** on the observed covariate values. A complementary approach, here implemented using stabilised inverse propensity weights (IPW), models the probability of belonging to each ethnic group given the same covariates, then uses those probabilities to reweight the observed searched population so that the covariate distributions are equalised across groups. This is most precisely described as **direct standardisation** of the searched population's covariate mix: because the propensity model is fitted on people who were already stopped (weighted by search count), the resulting rate ratios describe what the disparity would look like if all ethnic groups had the same covariate profile within the searched population, rather than providing a causal estimate in the general population sense.

Comparing GLM and IPW-standardised estimates is informative in two ways. First, if the two methods produce similar rate ratios, the findings are unlikely to depend on the parametric assumptions of the regression model. Second, where the estimates diverge, the divergence reveals something substantive: differences in how ethnic groups are distributed across high- and low-stop conditions.

All computations are implemented in `analysis/09_ipw.R`.

C.1 Method

Propensity model. A multinomial logistic regression is fitted to the modelling dataset, predicting ethnicity from the same five covariates used in the GLM: financial year, region, sex, age group, and reason for search. Each stratum is weighted by its search count so that the model estimates $\hat{P}(\text{ethnicity} \mid \text{covariates})$ within the searched population, not in the general population. The model converged in all cases.

Stabilised weights. For each stratum i with observed ethnicity g_i, the stabilised IPW weight is:

$$w_i = \frac{\hat{P}(E = g_i)}{\hat{P}(E = g_i \mid X_i)}$$

where $\hat{P}(E = g_i)$ is the marginal proportion of ethnicity g_i in the searched population and $\hat{P}(E = g_i \mid X_i)$ is the predicted probability from the propensity model. Stabilising by the

marginal probability reduces variance relative to unstabilised weights. Weights are trimmed at the 99th percentile (threshold: 19.4) to limit the influence of extreme values; 20 strata are affected by this trimming.

IPW rate ratios. Weighted search rates per 1,000 population are computed for each ethnic group and divided by the weighted White rate to give IPW rate ratios. Bootstrap percentile confidence intervals (B = 200) are obtained by resampling modelling strata with replacement and repeating the full weighting procedure.

C.2 Weight Distributions

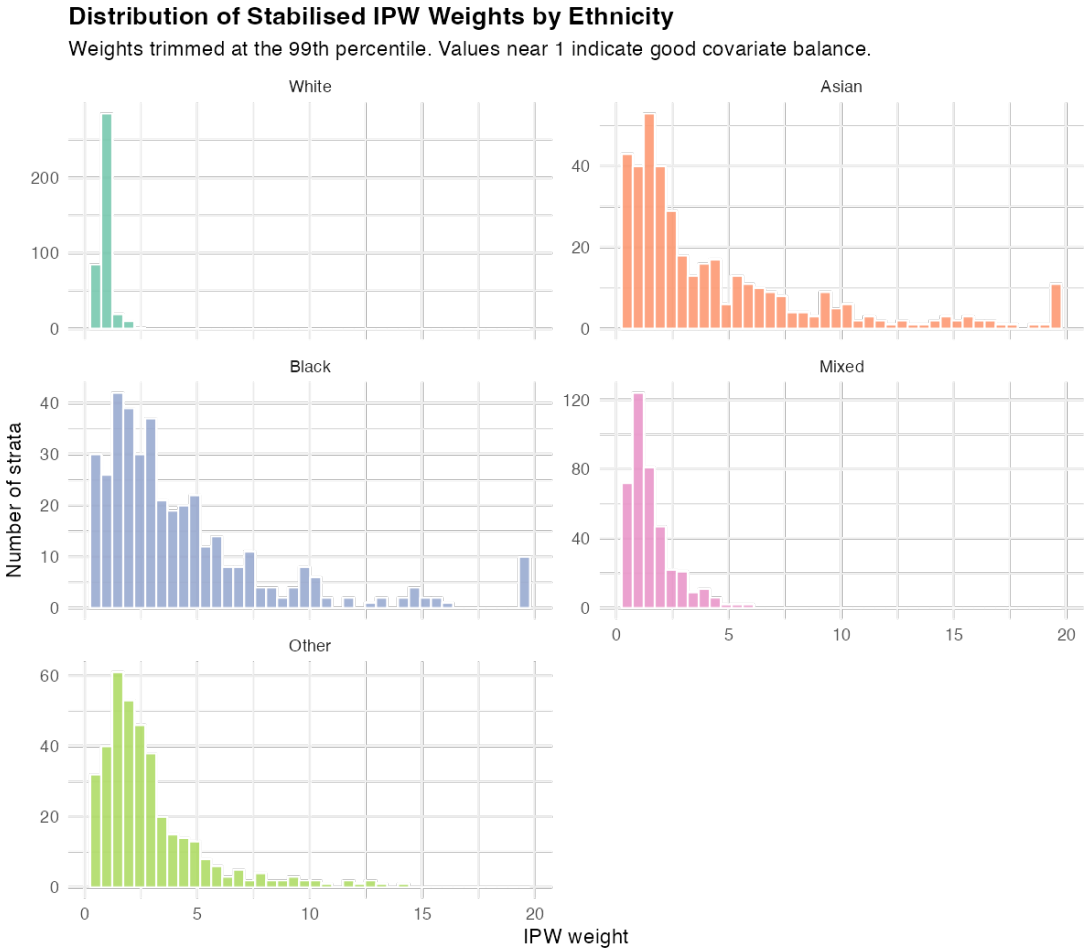


Figure C.1: Distribution of stabilised IPW weights by ethnicity group. White weights are highly concentrated near their mean (0.87), indicating that White people's covariate profile closely matches the overall searched population. Asian and Black weights are more dispersed and larger on average, reflecting the concentration of these groups in high-stop conditions such as London.

Figure C.1 shows the weight distributions by ethnicity. For White individuals, mean weight = 0.87, concentrated close to 1, which indicates that White people's observed covariate distribution closely resembles the overall searched population. For Black individuals, mean weight = 4.39, and for Asian individuals mean weight = 4.53. These larger weights reflect the fact that Black and Asian people are over-represented in conditions (particularly London, drug-related searches, younger age groups) where overall stop rates are high. The IPW procedure up-weights these individuals to represent what their rates would be if their covariate profile matched the full searched population.

The moderate extent of weight trimming (20 strata at the 99th percentile) suggests the propensity model achieves reasonable coverage without extreme extrapolation.

C.3 GLM vs IPW Comparison

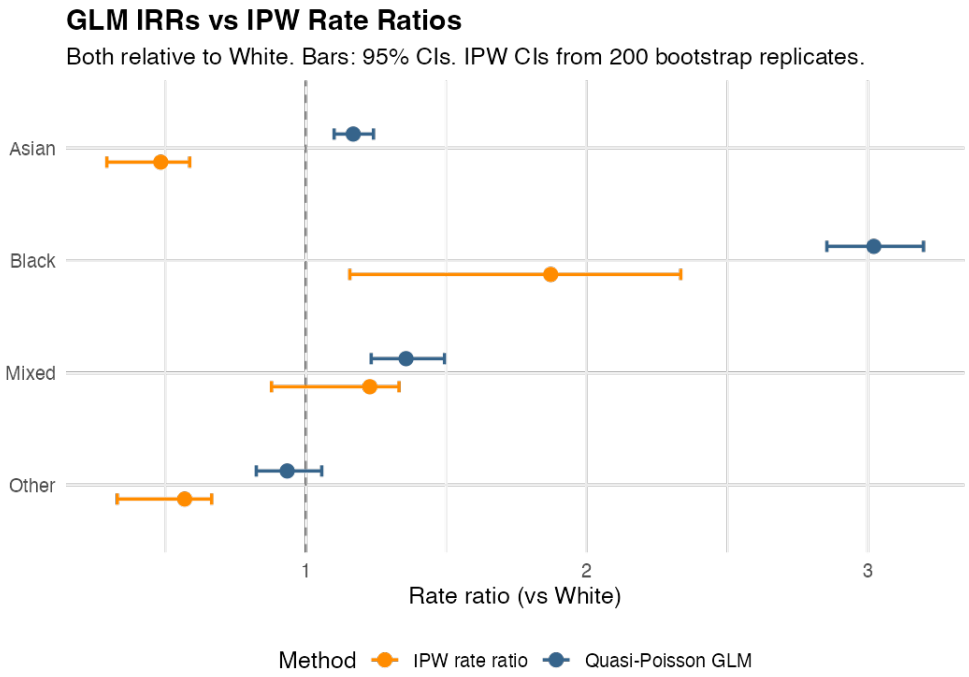


Figure C.2: GLM incidence rate ratios (blue) and IPW rate ratios (orange) for each non-White ethnic group, relative to White. Both estimates and their 95% confidence intervals are shown. The Black/White disparity is large under both methods. The Asian disparity reverses sign under IPW, indicating that the GLM Asian coefficient largely reflects covariate concentration rather than differential treatment.

Table C.1: GLM IRRs and IPW rate ratios with 95% confidence intervals. All estimates are relative to White. GLM CIs use the normal approximation; IPW CIs are percentile bootstrap intervals (B = 200).

	Quasi-Poisson GLM			IPW		
Ethnicity	GLM IRR	GLM CI lo	GLM CI hi	IPW RR	IPW CI lo	IPW CI hi
Black	3.022	2.855	3.199	1.872	1.157	2.335
Mixed	1.357	1.233	1.494	1.228	0.878	1.332
Asian	1.169	1.101	1.241	0.484	0.292	0.587
Other	0.934	0.824	1.057	0.569	0.329	0.665

Figure C.2 and Table C.1 present the side-by-side comparison. Three patterns are worth noting.

Black/White disparity: large under both methods. The GLM IRR is 3.022 (95% CI: 2.855, 3.199) and the IPW rate ratio is 1.872 (95% CI: 1.157, 2.335). The IPW estimate is lower, which is expected: the conditional GLM estimate asks "among searches of the same sex, age group and reason in the same region, how many Black searches are there per Black resident, relative to White searches per White resident?", while the IPW estimate asks "if Black and White people had the same overall covariate profile, what would the rate ratio be?" Because Black people are more concentrated in high-stop conditions (London, drug searches), the marginal IPW rate ratio is smaller than the conditional GLM IRR. Critically, both estimates are well above 1, and the IPW lower bound (1.157) excludes 1. The Black/White disparity is therefore robust to the choice of estimand and method.

Asian/White: divergence reveals geographic concentration. The GLM estimates a modest positive disparity (IRR = 1.169), but the IPW method produces a rate ratio of 0.484 (95% CI: 0.292, 0.587), well below 1. The mean Asian propensity weight of 4.53 is the largest of any group, indicating that the Asian-searched population is concentrated in conditions with high stop rates for all groups. Once the reweighting equalises those conditions, the Asian rate falls below the White rate. In other words, the positive GLM coefficient for Asian individuals reflects primarily their geographic and demographic concentration (particularly in London and in drug-related searches), not evidence of differential treatment conditional on being in similar circumstances. This does not imply the absence of racial discrimination in Asian stops, but it does show that the mechanism differs from the Black/White case.

Mixed and Other groups. The IPW rate ratios for Mixed (1.23, CI: 0.88, 1.33) and Other (0.57, CI: 0.33, 0.67) are somewhat lower than the corresponding GLM estimates, consistent with partial geographic concentration effects. The Mixed group shows a positive but uncertain IPW disparity. The Other group has lower rates than White under both methods.

C.4 Interpretation and Limitations

The GLM and IPW analyses are complementary rather than competing. The GLM IRR is the appropriate summary for the question "within comparable conditions, are Black individuals stopped more than White individuals?", to which the answer is yes, by a factor of approximately 3. The IPW rate ratio addresses "across the whole searched population, standardised to a common covariate profile, what is the disparity?", to which the answer is also yes, by a factor of approximately 1.9. Both are valid and both confirm a substantial Black/White disparity.

Three limitations apply to this analysis. First, because the propensity model is fitted on the searched population weighted by count, the estimates are more accurately characterised as direct standardisation within the searched population than as classical causal IPW in the general population. A causal interpretation would additionally require an unconfoundedness assumption holding in the full population, which cannot be verified here. Second, the analysis relies on the propensity model being correctly specified within the searched population; it uses the same covariates as the GLM and has no access to unmeasured variables such as police force policies or local crime conditions. Third, the large weights for Asian and Black groups (mean 4.5 and 4.4 respectively) indicate limited covariate overlap, which increases the variance of the standardised estimates and means the bootstrap confidence intervals for these groups are relatively wide.